

Periods and L -functions of meromorphic modular forms via finite contours

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Abstract

In this note, we synthesize and generalize previous work, which (1) extracted periods for weak cusp forms in a manifestly finite manner [1], (2) identified a non-holomorphic regularization prescription to define finite analogs of L-functions associated to weakly holomorphic modular forms [2], and (3) identified a contour prescription to define an L-function for meromorphic modular forms with poles inside the fundamental domain [3]. Our construction, derived from the cohomological period extraction of weak cusp forms in [1], shows that these three lines of work not only naturally align, but derive from the *same* manifestly finite linear functional acting on the space of $\mathrm{SL}_2(\mathbb{Z})$ modular forms.

We call this linear functional the L -integral, and denote it by $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$, where $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ are paths from $S^{-1}\tau_0, T^{-1}\tau_0$, respectively, to τ_0 . When $f \in M_k$, $L^*(f, s) = \Gamma(s) \sum_{n \neq 0} \frac{c_f(n)}{(2\pi n)^s}$: it is *the* classical L-function. Our L-integral is finite and τ_0 -independent (within each admissible contour homotopy class) for any meromorphic modular form f .

We use this L-integral to (1) extract published numerical values of the periods of the weak cusp form $\widehat{\Delta} := \frac{1}{q} + O(q^2) \in S_{12}^1$ in [1], (2) recover the regularized expressions for the divergent Mellin transform of f in [2], and (3) recover the L-functions of [3] for meromorphic forms with interior poles that are regular at the cusps. When f has poles inside the fundamental domain, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is unique and constant for paths within the same homotopy class; related $L^*(f, s)$ -integrals along paths in different homotopy classes differ by wall-crossing residues. Finally, all major results were tested to arbitrary (50 digit) precision in our python scripts [4].

1 Introduction and main results

In this paper, we introduce a general framework that defines L-functions for generic modular forms on the full group $\mathrm{SL}_2(\mathbb{Z})$, and extracts their periods for the explicit case when the holomorphic cusp forms are one-dimensional, i.e. when $\dim(S_k) = 1$. Classically, L-functions and periods are both well-defined and well-studied for holomorphic modular forms. However, there is a growing body of work exploring periods [1] and L-functions [2] for weakly holomorphic modular forms. Similarly, there are natural physical motivations to consider L-functions for properly meromorphic modular forms, which have poles *within* the fundamental domain [3].

In these works, and specifically in [2, 3], and throughout physics, divergences due to singularities in quantities of interest necessitate the introduction of regulators in order to define finite quantities. Regulators, though, often break symmetries, and care is needed to check that the regulator has not injected undue violence to the system. Of relevance here is that [2] introduced a non-holomorphic regulator to deal with the singularities at either end of the classical Mellin contour, $\int_0^\infty$, sufficient to regulate divergences at the cusps, but which violates holomorphy in the deformed contour-integral, ruining path-independence. Simultaneously, [3] concerned itself with L-functions for modular forms which diverge within the interior of the fundamental domain. Here, holomorphic dependence on τ is paramount, which puts this treatment at odds with the regulator in [2].

In this note, we introduce manifestly finite contour integral functionals to define L-functions for generic meromorphic modular forms that are of sufficient generality to handle any combination of poles at cusps and poles within the fundamental domain. Our construction is motivated by, and makes precise contact with, work which extracts period-coefficients from weak cusp forms [1]. Our integrals are manifestly finite throughout, and reproduce all results in the literature, unifying the quasi-period results and the L-function extensions within one concrete, natural framework.

What we construct. Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, let f be a meromorphic modular form of even weight $k \in \mathbb{Z}$ on Γ , and let $\tau_0 \in \mathbb{H}$ be a basepoint with the property that the pair of paths

$$\begin{cases} S\text{-segment:} & \gamma_{\tau_0}^S : [0, 1] \rightarrow \mathbb{H}, & \gamma_{\tau_0}^S(0) = -1/\tau_0, \gamma_{\tau_0}^S(1) = \tau_0, \\ T\text{-segment:} & \gamma_{\tau_0}^T : [0, 1] \rightarrow \mathbb{H}, & \gamma_{\tau_0}^T(0) = \tau_0 - 1, \gamma_{\tau_0}^T(1) = \tau_0, \end{cases} \quad (1)$$

in $\mathbb{H}$ avoids the Γ -orbit of the poles of f in $\mathbb{H}$. Using these contours, we define

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right], \quad (2)$$

where the T -kernel is built from the following explicit two Hurwitz-zeta functions:

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1). \quad (3)$$

When the contour avoids poles in f , both contour integrals contributing to $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ in (2) are finite for every basepoint $\tau_0 \in \mathbb{H}$ admissible for f and every $s \in \mathbb{C}$; no regulator, asymptotic-subtraction argument, or analytic-continuation step is required. Further, as we will show, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is invariant under both τ_0 -variations and $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ -variations, as long as neither $\gamma_{\tau_0}^S$ nor $\gamma_{\tau_0}^T$ cross any pole location within the fundamental domain.

We will show that $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is the classical L-function for $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$. Further, because of its τ_0 -independence and finiteness for $f \in M_k^!(\mathrm{SL}_2(\mathbb{Z}))$, and because we reproduce the analytic formulae in [2], we unambiguously show that their candidate L-function is the canonical L-function one would associate with any weak modular form, $f \in M_k^!$.

Three theorems. The functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ obeys three properties that together identify it as the canonical finite-contour L-function of f :

- (i) (*Theorem 1.1*) When $\dim S_k = 1$, the integer- s specialisations of (2) reduce to a polynomial-kernel form that produces the periods $\omega^\pm(f)$ and quasi-periods $\eta^\pm(f)$ as an explicit linear functional in the Fourier coefficients of f . For $f = \Delta$ at $k = 12$ this recovers the Brown values of $\omega^\pm(\Delta)$; for $f = \hat{\Delta}$ this gives the quasi-periods $\eta^\pm(\hat{\Delta})$ from a finite-endpoint contour pair, with no divergence at the cusp.
- (ii) (*Theorem 1.2*) For $f \in M_k^!$, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is τ_0 -independent on $\mathbb{H}$ and produces the completed L-function $L^*(f, s)$ at every $s \in \mathbb{C}$ as a single closed-form incomplete-gamma sum, agreeing term by term in the Fourier expansion with [2]. Our proof in fact exploits the simplicity at the elliptic fixed-point basepoint $\tau_0 = i$, where the S -segment collapses to a point and (2) reduces to a single integral on the compact contour $[i-1, i]$.
- (iii) (*Theorem 1.3*) For general meromorphic f , $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends on $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ only through their homotopy class in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, with explicit wall-crossing residue jumps when the homotopy class changes by crossing a pole. Restricted to forms regular at the cusp, this reproduces the deformed-Mellin L-function $L_f^*(s)_{\mathrm{int}}$ (Corollary 1.4; see [3]).

The complete statements are below; their proofs occupy §3–§5.

Theorem 1.1 (Periods and quasi-periods as an explicit linear functional, $\dim S_k \leq 1$). *Let $k \in 2\mathbb{Z}$ with $\dim S_k \leq 1$, and let $f \in S_k^!$. For the substantive cases $k \in \{12, 16, 18, 20, 22, 26\}$ where $\dim S_k = 1$, the period and quasi-period coefficients $\omega^\pm(f)$ are given by the explicit two-segment linear functional*

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right] \quad (4)$$

for every $f \in S_k^!$, every $\tau_0 \in \mathbb{H}$, and every interior $\ell \in \{2, \dots, k-2\}$, with parity $\pm$ determined by $\ell - 1 \bmod 2$. Here $c_S^{\ell-1} := [p_f^\pm]_{X^{n-(\ell-1)} Y^{\ell-1}} \in \mathbb{Q}$ is the Brown cuspidal-basis coefficient at the matched monomial ([1, 12]), $\tilde{k}_T(\tau, s)$ is the Hurwitz-zeta T -kernel of (3), and its integer- $s = \ell$ specialisation collapses via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ ($m \geq 0$) to the $\mathbb{Q}$ -rational polynomial

$$\tilde{k}_T(\tau, \ell) = -\frac{B_\ell(\tau+1)}{\ell} + (-1)^{\ell-1} \frac{B_{k-\ell}(\tau+1)}{k-\ell} \in \mathbb{Q}[\tau] \quad (5)$$

of degree at most $k-2$. The right side of (4) is independent of $\tau_0 \in \mathbb{H}$.

Inside the two-segment bracket of (4) sit only the bare Mellin monomial $\tau^{\ell-1}$ on $\gamma_{\tau_0}^S$ and the bare Hurwitz polynomial $\tilde{k}_T(\tau, \ell)$ on $\gamma_{\tau_0}^T$; the factors $(2\pi i)^{k-1}$, $(-1)^{\ell-1}$, $\binom{k-2}{\ell-1}$ and the Brown basis coefficient $c_S^{\ell-1}$ are absorbed into the prefactor.

Higher-dimensional S_k . For $k \geq 24$ with $\dim S_k \geq 2$, the Brown cuspidal-basis coefficient $c_S^{\ell-1}$ in (4) becomes matrix-valued: each $\ell \in \{2, \dots, k-2\}$ gives a $\dim S_k$ -component linear functional on $\{c_f(n)\}$ rather than a scalar. The structural argument behind Theorem 1.1 — coboundary inversion on V_{k-2} via δ_T^{-1} in closed Bernoulli-operator form — generalises without conceptual change, but the explicit kernel tables grow with $\dim S_k$, and the ℓ - and τ_0 -independence statement of Corollary 3.8 becomes a matrix identity. Working out the explicit matrix-valued kernels for $k = 24, 28, 30, \dots$ and connecting them to the existing Eichler–Shimura matrix-period basis literature is the natural generalisation of Theorem 1.1, and we leave it to a follow-up. Theorems 1.2 and 1.3 are unaffected and apply at every even k .

Theorem 1.2 (L -function for weakly holomorphic forms). *Let $k \in 2\mathbb{Z}$ and $f \in M_k^!$. With $\tilde{k}_T(\tau, s)$ the Hurwitz-zeta kernel of (3), for every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right] \quad (6)$$

is finite and independent of both $\tau_0 \in \mathbb{H}$ and of the admissible contour choices $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$. We write $L^*(f, s)$ for the common value.

Specialising to the elliptic fixed point $\tau_0 = i$ of the modular involution $S: \tau \mapsto -1/\tau$, the S -segment $[-1/i, i]$ collapses to the single point i , the S -integral in (6) vanishes, and L^* reduces to a single integral on the compact contour $[i-1, i] \subset \mathbb{H}$, with closed-form sum

$$L^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right]. \quad (7)$$

The right side of (7) is the completed L -function $L^*(f, s)$ of f . For holomorphic modular forms it reduces to the classical Hecke–Weil Mellin transform. For weak modular forms, it reduces to the results of [2], proving that their non-holomorphic regulator prescription gives the unique, canonical L -function for every weak modular form.

Classical Mellin recovery and canonical uniqueness. For $f \in S_k$ a holomorphic cusp form, taking the cusp-limit $\tau_0 \rightarrow i\infty$ in (6) collapses the S -segment onto the classical Mellin contour $[0, i\infty]$ while the T -segment vanishes exponentially (since $f(\tau) = O(e^{-2\pi \operatorname{Im} \tau})$ on the segment as $\operatorname{Im} \tau \rightarrow \infty$):

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt, \quad (8)$$

the classical completed L -integral of f . For $f \in M_k$ with $c_f(0) \neq 0$ the same limit holds after a standard constant-term subtraction (see [3, Lemma 2.1] for the explicit polynomial correction).

For $f \in M_k^!$ with cusp poles the cusp-limit is genuinely pathological: $f(\tau) \rightarrow \infty$ as $\operatorname{Im} \tau \rightarrow \infty$, so the S -segment integral diverges along the imaginary axis, and the T -segment integral $\int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau$ acquires a matching divergence of the opposite sign as $\operatorname{Im} \tau_0 \rightarrow \infty$ — made precise in Lemma 5.4, where the T -segment is shown to equal an explicit incomplete-gamma sum whose polar mode q^{-M} contributes a term growing like $|c_f(-M)| e^{2\pi M \operatorname{Im}(\tau_0)}$. The two divergences cancel at every finite τ_0 , so $L^*(f, s)$ is finite and well defined throughout $\mathbb{H}$, but no naive Mellin contour limit exists. The construction of [2] recovers $L^*(f, s)$ on $M_k^!$ by a non-holomorphic regulator at the cusps; Theorem 1.2 shows that this regulator and the finite-contour functional (6) compute the same object, term by term in the Fourier expansion. In particular, $L^*(f, s)$ is independent of any choice of regularisation: it is the value of the finite-contour functional at any admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$, and the BFK construction is one closed-form evaluation of it (at $\tau_0 = i$).

Theorem 1.3 (*L -function for meromorphic modular forms*). *Let $k \in 2\mathbb{Z}$ and let f be a meromorphic modular form of weight k on Γ . Write $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ for the upper half-plane with the Γ -orbit of the interior poles of f removed, and call the pair $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ admissible if $\tau_0 \in \mathbb{H}_f$ and both contours lie in $\mathbb{H}_f$, with $\gamma_{\tau_0}^S$ running from $S^{-1}\tau_0 = -1/\tau_0$ to τ_0 and $\gamma_{\tau_0}^T$ from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 . Define $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ by (2) with the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ of (3). Then:*

- (a) *Both integrals in (2) are finite for every admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and every $s \in \mathbb{C}$.*
- (b) *$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends on the admissible pair only through its homotopy class in $\mathbb{H}_f$: any continuous one-parameter family of admissible pairs — including variations of the anchoring basepoint τ_0 — leaves the value unchanged.*
- (c) *If two admissible configurations $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and $(\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$ lie in homotopy classes related by a single crossing of an interior pole $\tau_p \in \mathbb{H}$ of f , then*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) - L^*(f, s; \gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T) = \mathcal{R}_{\tau_p}(f, s), \quad (9)$$

where for a simple pole $\operatorname{Res}(f, \tau_p)$,

$$\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \operatorname{Res}(f, \tau_p) \left[\sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (10)$$

with $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$ the winding-number contributions of $\gamma_{\tau_0}^S$ and $\gamma_{\tau_0}^T$ across τ_p (the generic case has one nonzero and the other zero, according to which contour crossed). Higher-order poles contribute via the corresponding derivatives in τ_p of (10); the closed form is given in §5.

Corollary 1.4 (L^* on regular-at-cusps meromorphic forms). *Let $f \in F_k$ be a meromorphic modular form of weight k that is regular at the cusp. Let $L_f^*(s)_{\text{int}, \gamma}$ denote the deformed-Mellin L -function of f along a contour γ from 0 to $i\infty$ lying in $\mathbb{H}_f$ (see [3, §2]). Then for every admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ whose homotopy class in $\mathbb{H}_f$ is compatible with γ (in the sense that the contour $\gamma_{\tau_0}^S$ together with the cusp-limit piece $\lim_{\tau_0 \rightarrow i\infty} \gamma_{\tau_0}^S \rightarrow [0, i\infty]$ can be deformed to γ in $\mathbb{H}_f$),*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_f^*(s)_{\text{int}, \gamma} \quad \text{for every } s \in \mathbb{C}. \quad (11)$$

The structural picture. The contour pair (1) has two natural one-parameter specialisations of the basepoint, each exploiting a fixed point of the modular curve $Y(1)$: the cusp specialisation $\tau_0 \rightarrow i\infty$ sends the S -segment to the classical Mellin contour $[0, i\infty]$ and the T -segment to 0 exponentially; the elliptic specialisation $\tau_0 = i$ kills the S -segment identically and leaves L^* as a single integral on $[i-1, i]$. The cusp specialisation, applied at finite $\tau_0 = i\Lambda$, gives the Λ -truncated Mellin contour of $L_f^*(s)_{\text{int}}$ (see [3]) as the S -segment alone, with the T -segment serving as the missing ingredient that produces the cocycle and renders the $\Lambda \rightarrow \infty$ limit unnecessary. The Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ in (3) is the closed form of the T -segment kernel: it is the unique V_n -analog, valid at every $s \in \mathbb{C}$, of the polynomial coboundary inversion of [7] (which itself recovers as the integer- s specialisation, by Theorem 1.1).

Outline. §2 introduces the period-defining cocycle and its moment expansion in the polynomial coefficient system V_{k-2} . §3 identifies the bare Mellin and Hurwitz integer- s specialisations as the period-extracting kernels and proves Theorem 1.1. §4 derives the Hurwitz-zeta kernel $\tilde{k}_T$ via a coboundary equation, proves Theorem 1.2 via a Hurwitz-unfolding lemma, and contains an explicit worked example at $k = 12$ exhibiting the Bernoulli-polynomial collapse of $\tilde{k}_T$ at integer s together with numerical values of $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$. §5 proves Theorem 1.3 and Corollary 1.4. §6 collects conceptual framings (holonomy/Wilson-loop interpretation, on-shell-amplitudes analogy, the regularisation-as-symptom heuristic, the open question on the geometric content of the $\tau^{s-1} f d\tau$ Mellin kernel, and pointers to the sequel on systematic per-mode theory for interior-pole forms). Appendix A records the algebraic derivation of the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ from contour-invariance of the two-segment functional: starting from the classical Mellin transform split at the contour endpoints $\{-1/\tau_0, \tau_0\}$ and demanding that the resulting incomplete-gamma tail be representable as a T -segment integral, one derives a shift-difference equation whose solution is the Hurwitz-zeta combination of (3). Appendix B is a brief Eichler–Shimura primer for readers approaching the subject from theoretical physics. This appendix is included for the working-version of this note and is intended to be removed before formal publication.

2 The period-defining cocycle and its moment expansion

The polynomial coefficient system. Let $V_n := \mathbb{C}[X, Y]_n$ be the space of complex homogeneous polynomials in two auxiliary variables X, Y of degree n , equipped with the right Γ -action

$$(P|_\gamma)(X, Y) = P(aX + bY, cX + dY) \quad \text{for } \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma.$$

For a meromorphic modular form f of weight k , write $n := k-2$ throughout and form the V_n -valued 1-form on $\mathbb{H}$

$$\omega_f(\tau) := f(\tau) (X - \tau Y)^n d\tau.$$

Modularity of f implies that ω_f is Γ -equivariant in the standard sense [12]: under $\tau \mapsto \gamma^{-1}\tau$, the weight- k transformation of f cancels the weight- $(-n)$ slash on $(X - \tau Y)^n$, so the V_n -valued 1-form ω_f is invariant under the simultaneous action of Γ on τ and on the auxiliary variables (X, Y) .

The construction of ω_f above uses only the weight- k modularity of f and makes no cuspidality assumption: ω_f is well-defined for every meromorphic modular form $f \in M_k^!$ (and indeed for meromorphic forms with poles in the fundamental domain, provided contours avoid the Γ -orbit of those poles), with the cuspidal hypothesis $f \in S_k^!$ entering only later via Lemma 3.1(c) as the solvability condition for the period-polynomial coboundary equation.

The cocycle. The two-segment cocycle construction used in the present note is the standard Eichler–Shimura cohomological setup [12], in which a weakly holomorphic modular form is integrated against the polynomial coefficient system $(X - \tau Y)^n$ along contours from $\gamma^{-1}\tau_0$ to τ_0 for the generators $\gamma \in \{S, T\}$ of Γ , with periods and quasi-periods extracted from the resulting 1-cocycle modulo coboundaries. The author was led to this presentation, and to its quasi-period reformulation, through personal correspondence with F. Brown [5]. Concretely: fix a basepoint $\tau_0 \in \mathbb{H}$ not on the Γ -orbit of any pole of f , and define

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f = (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^n d\tau \in V_n \otimes \mathbb{C}, \quad (12)$$

the contour being any path from $\gamma^{-1}\tau_0$ to τ_0 in $\mathbb{H}$ avoiding the Γ -orbit of the poles of f . The integral is independent of the choice of path within the same homotopy class in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$.

Lemma 2.1 (Cocycle property). *For all $g, h \in \Gamma$,*

$$C_{gh}^f(\tau_0) = C_g^f(\tau_0)|_h + C_h^f(\tau_0). \quad (13)$$

Equivalently, the map $\gamma \mapsto C_\gamma^f(\tau_0)$ is a 1-cocycle of Γ with coefficients in $V_n \otimes \mathbb{C}$.

Proof. Using $(gh)^{-1} = h^{-1}g^{-1}$ and path-additivity through the intermediate point $h^{-1}\tau_0$,

$$\int_{(gh)^{-1}\tau_0}^{\tau_0} = \int_{h^{-1}g^{-1}\tau_0}^{\tau_0} = \int_{h^{-1}g^{-1}\tau_0}^{h^{-1}\tau_0} + \int_{h^{-1}\tau_0}^{\tau_0}.$$

In the first integral on the right substitute $\tau = h^{-1}\sigma$, so σ runs from $g^{-1}\tau_0$ to τ_0 . The Γ -equivariance of ω_f converts the integrand to $\omega_f(\sigma)|_h$, yielding the first summand of (13). The second integral gives the second summand directly. $\square$

A cocycle on $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ is determined by its values on the two generators

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

subject to the relations $S^2 = (ST)^3 = -I$ (which act trivially on V_n for n even). We write

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \int_{-1/\tau_0}^{\tau_0} \omega_f, \quad C_T^f(\tau_0) = (2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} \omega_f, \quad (14)$$

since $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$. These are the two integrals over the contour pair (1).

Moment expansion. Expanding $(X - \tau Y)^n = \sum_{\ell=0}^n \binom{n}{\ell} (-\tau)^\ell X^{n-\ell} Y^\ell$ and pulling the polynomial out of the integral defines the moment integrals

$$I_S^\ell(f, \tau_0) := \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^\ell d\tau, \quad I_T^\ell(f, \tau_0) := \int_{\tau_0-1}^{\tau_0} f(\tau) \tau^\ell d\tau, \quad (15)$$

for $\ell = 0, 1, \dots, n$, in terms of which

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell I_S^\ell(f, \tau_0) X^{n-\ell} Y^\ell,$$

and similarly for $C_T^f(\tau_0)$. Theorem 1.1 extracts $\omega^\pm(f)$ from particular linear combinations of these moments.

Basepoint dependence. Varying τ_0 changes C^f by a coboundary, provided f has no interior poles. Define an antiderivative $F(\tau)$ of ω_f on a simply connected open subset of $\mathbb{H}$ avoiding $\Gamma \cdot f^{-1}(\infty)$. Then for any two basepoints τ_0, τ'_0 in the same connected component, path-additivity gives

$$C_\gamma^f(\tau'_0) - C_\gamma^f(\tau_0) = (P|_\gamma - P) \quad \text{with} \quad P = (2\pi i)^{n+1} (F(\tau'_0) - F(\tau_0)) \in V_n \otimes \mathbb{C}. \quad (16)$$

This is the coboundary $\delta P(\gamma) := P|_\gamma - P$. When f has interior poles, dragging τ_0 across a pole picks up a Cauchy residue contribution beyond (16); this wall-crossing structure is the content of Theorem 1.3 in §5.

Cohomology and periods. The cohomology class $[C^f] \in H^1(\Gamma; V_n)$ is independent of τ_0 (on the component) and independent of the choice of δP representative. The classical Eichler–Shimura theorem [12] identifies the cuspidal subspace $H_{\text{cusp}}^1(\Gamma; V_n)$ with $S_k \oplus \overline{S_k}$, and the linear coordinates of $[C^f]$ in this cuspidal subspace are the periods and quasi-periods $\omega^\pm(f), \eta^\pm(f)$ [1].

3 Period extraction: proof of Theorem 1.1

Set $n := k - 2$ throughout. The argument is k -uniform on the $\dim S_k = 1$ family $k \in \{12, 16, 18, 20, 22, 26\}$: every step below — the cocycle structure of §2, the δ_T -solvability of Lemma 3.1, the Bernoulli closed-form inversion of Lemma 3.2, the Hurwitz-zeta kernel identification of Proposition 3.6, and the ℓ - and τ_0 -independence of Corollary 3.8 — holds for every even k with the Brown cuspidal-basis coefficient $c_S^{\ell-1}$ replaced by the relevant weight's basis-polynomial coefficient. The lower even weights $k \in \{2, 4, 6, 8, 10, 14\}$ in the $\dim S_k \leq 1$ family have $S_k = 0$, so the theorem is vacuous there. Numerical verification at $k = 12$ against Brown's published values for the canonical pair $f \in \{\Delta, \hat{\Delta}\}$ is consolidated in §3.1.

Lemma 3.1 (Solvability of $\delta_T P = C_T^f$ is cuspidality of f). *Define the linear endomorphism $\delta_T: V_n \rightarrow V_n$ by $\delta_T(P) := P|_T - P$, where $T \in \Gamma$ acts on V_n as $(X, Y) \mapsto (X + Y, Y)$. Then:*

- (a) $\ker(\delta_T) = \mathbb{C} \cdot Y^n$.
- (b) $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$.

(c) For any meromorphic modular form f of weight $k = n + 2$ and any $\tau_0 \in \mathbb{H}$ admissible for f ,

$$[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} \cdot c_f(0), \quad (17)$$

where $c_f(0)$ is the constant Fourier coefficient of f .

Consequently, the equation $\delta_T P = C_T^f(\tau_0)$ is solvable for $P \in V_n$ if and only if $c_f(0) = 0$, i.e. if and only if f is a (weak) cusp form. When solvable, P is unique modulo $\mathbb{C} \cdot Y^n$.

Proof. (a) T acts on V_n as the translation $X \mapsto X + Y$, fixing only powers of Y . Hence T -invariants in V_n , equivalently $\ker(\delta_T)$, are exactly $\mathbb{C} \cdot Y^n$.

(b) Writing $P = \sum_{\ell=0}^n p_\ell X^{n-\ell} Y^\ell$ and expanding $(X + Y)^{n-\ell}$ binomially,

$$[\delta_T P]_{X^{n-m} Y^m} = \sum_{\ell=0}^{m-1} \binom{n-\ell}{m-\ell} p_\ell \quad (m = 0, 1, \dots, n). \quad (18)$$

This is strictly lower-triangular in the ordered basis $(X^n, X^{n-1}Y, \dots, Y^n)$: the $m = 0$ row is the empty sum, identically zero, and the off-diagonal entries $\binom{n-\ell}{m-\ell}$ for $\ell < m$ are positive. The $m = 0$ row forces $[\delta_T P]_{X^n} = 0$ for every P , while rows $m = 1, \dots, n$ determine $p_0, \dots, p_{n-1}$ uniquely from any prescribed right-hand side, with p_n unconstrained. Hence $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$, and the solution P is unique modulo $\mathbb{C} \cdot Y^n$.

(c) The moment expansion (14) gives $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} I_T^0(f, \tau_0)$ where $I_T^0(f, \tau_0) = \int_{\tau_0-1}^{\tau_0} f(\tau) d\tau$. Substituting the Fourier expansion $f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau}$, absolutely convergent on the compact T -segment, and integrating term by term,

$$\int_{\tau_0-1}^{\tau_0} e^{2\pi i n \tau} d\tau = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n \neq 0, \end{cases}$$

by T -periodicity over a unit segment. Hence $I_T^0(f, \tau_0) = c_f(0)$ for every admissible τ_0 , which proves (17).

Combining (b) and (c) gives the solvability statement. $\square$

Lemma 3.2 (Bernoulli closed form for δ_T^{-1}). *For every $Q \in V_n$ with vanishing X^n -coefficient there exists $P \in V_n$ unique modulo $\mathbb{C} \cdot Y^n$ with $\delta_T P = Q$. Writing $Q = \sum_{r=1}^n q_r X^{n-r} Y^r$ and $P = \sum_{m=0}^n p_m X^{n-m} Y^m$, the solution is given in closed form by*

$$p_m = \sum_{r=1}^{m+1} \frac{B_{m-r+1} (n-r)!}{(m-r+1)! (n-m)!} q_r \quad (m = 0, 1, \dots, n-1), \quad (19)$$

where B_ℓ are Bernoulli numbers ($B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_{2\ell+1} = 0$ for $\ell \geq 1$); the coefficient p_n is unconstrained and parametrises $\ker(\delta_T) = \mathbb{C} \cdot Y^n$.

Proof. The operator identity $T = e^{Y\partial_X}$ on V_n (since T acts as the translation $X \mapsto X + Y$ by Y) gives $\delta_T = e^{Y\partial_X} - 1$. The operator $Y\partial_X$ is nilpotent on V_n : each application reduces the X -degree by one, so $(Y\partial_X)^{n+1} = 0$ on V_n . Consequently, the formal power series in $Y\partial_X$ obtained from the Bernoulli generating function $z/(e^z - 1) = \sum_{k \geq 0} (B_k/k!) z^k$ truncates to a finite $\mathbb{Q}$ -linear combination of operators on V_n , and supplies the closed-form inverse

$$\delta_T^{-1} = (Y\partial_X)^{-1} \frac{Y\partial_X}{e^{Y\partial_X} - 1} = \sum_{k \geq 0} \frac{B_k}{k!} (Y\partial_X)^{k-1}, \quad (20)$$

in which the $k = 0$ summand is the formal antiderivative $(Y\partial_X)^{-1}$, acting on monomials by $(Y\partial_X)^{-1}(X^a Y^b) = X^{a+1} Y^b / (a+1)$ (for $b \geq 1$), and the $k \geq 1$ summands are differential operators that eventually annihilate any polynomial of bounded X -degree. Applied to $X^{n-r} Y^r$ for $r \geq 1$,

$$\delta_T^{-1}(X^{n-r} Y^r) = \sum_{k=0}^{n-r+1} \frac{B_k}{k!} \frac{(n-r)!}{(n-r-k+1)!} X^{n-r-k+1} Y^{r+k-1}. \quad (21)$$

Summing $Q = \sum_r q_r X^{n-r} Y^r$ termwise and reading off the $X^{n-m} Y^m$ coefficient identifies $k = m-r+1$ (so r runs over $\{1, \dots, m+1\}$, k over $\{0, \dots, m\}$), yielding (19). Existence and uniqueness modulo $\mathbb{C} \cdot Y^n$ follow from Lemma 3.1(a, b). $\square$

Definition 3.3 (Bernoulli T -segment kernel $\widehat{k}_T(\ell, \tau)$). *For non-negative integers l with $l \leq n-1$, set the $\mathbb{Q}$ -rational coefficients*

$$\beta_{l,r} := \frac{(-1)^r \binom{n}{r} B_{l-r+1} (n-r)!}{(l-r+1)! (n-l)!} \quad \text{for } 1 \leq r \leq l+1, \quad (22)$$

and $\beta_{l,r} := 0$ for r outside $\{1, \dots, l+1\}$. For each interior $\ell \in \{2, \dots, k-2\}$, define the $\mathbb{Q}$ -rational polynomial

$$\widehat{k}_T(\ell, \tau) := \sum_{r=1}^n \left[\beta_{\ell-1,r} - (-1)^{\ell-1} \beta_{n-\ell+1,r} \right] \tau^r \in \mathbb{Q}[\tau], \quad (23)$$

of degree $\leq n$.

Lemma 3.4 (Integer- s collapse of the Hurwitz-zeta T -kernel). *For each integer $\ell \in \{2, \dots, k-2\}$, the Hurwitz-zeta T -kernel $\widetilde{k}_T(\tau, s)$ of (3) specialised to $s = \ell$ collapses to the $\mathbb{Q}$ -rational polynomial*

$$\widetilde{k}_T(\tau, \ell) = -\frac{B_\ell(\tau+1)}{\ell} + (-1)^{\ell-1} \frac{B_{k-\ell}(\tau+1)}{k-\ell} \in \mathbb{Q}[\tau] \quad (24)$$

of degree at most $k-2$.

Proof. The Hurwitz-zeta collapse $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ for non-negative integer m , applied at $m = \ell-1$ and $m = k-\ell-1$ (both non-negative for $\ell \in \{2, \dots, k-2\}$), together with $e^{i\pi(\ell-1)} = (-1)^{\ell-1}$, converts (3) at $s = \ell$ to (24). $\square$

Lemma 3.5 (Bernoulli-form vs. Hurwitz-form T -kernel: polynomial identity and integral equivalence). *For each interior $\ell \in \{2, \dots, k-2\}$, the Bernoulli T -kernel $\widehat{k}_T(\ell, \tau)$ of (23) and the Hurwitz T -kernel $\widetilde{k}_T(\tau, \ell)$ of (24) are related by the polynomial identity*

$$\widehat{k}_T(\ell, \tau) = (-1)^{\ell-1} \binom{n}{\ell-1} \left[\widetilde{k}_T(\tau, \ell) - \widetilde{k}_T(0, \ell) \right] \in \mathbb{Q}[\tau]. \quad (25)$$

Equivalently, the difference $\widehat{k}_T(\ell, \tau) - (-1)^{\ell-1} \binom{n}{\ell-1} \widetilde{k}_T(\tau, \ell) = -(-1)^{\ell-1} \binom{n}{\ell-1} \widetilde{k}_T(0, \ell) \in \mathbb{Q}$ is a constant. Consequently, for every $f \in S_k^!$ and every $\tau_0 \in \mathbb{H}$,

$$\int_{\gamma_{\tau_0}^T} f(\tau) \widehat{k}_T(\ell, \tau) d\tau = (-1)^{\ell-1} \binom{n}{\ell-1} \int_{\gamma_{\tau_0}^T} f(\tau) \widetilde{k}_T(\tau, \ell) d\tau. \quad (26)$$

Proof. Polynomial identity. Both sides of (25) are $\mathbb{Q}$ -rational polynomials in τ of degree $\leq k-2$, and both have vanishing constant term: the left side by construction ((23) sums from $r=1$), and the right side because $\tilde{k}_T(\tau, \ell) - \tilde{k}_T(0, \ell)$ vanishes at $\tau=0$. By direct comparison of the τ^r coefficients for $1 \leq r \leq k-2$ using (23) for the left side and (24) for the right side, together with the standard Bernoulli polynomial expansion $B_n(\tau+1) = \sum_{s=0}^n \binom{n}{s} B_s \tau^{n-s} + n \tau^{n-1}$, the two polynomial coefficients agree. Hence the polynomial identity (25) holds in $\mathbb{Q}[\tau]$.

Integral equivalence. The difference $\tilde{k}_T(\ell, \tau) - (-1)^{\ell-1} \binom{n}{\ell-1} \tilde{k}_T(\tau, \ell) = -(-1)^{\ell-1} \binom{n}{\ell-1} \tilde{k}_T(0, \ell) \in \mathbb{Q}$ is constant by (25). For $f \in S_k^!$ the integral of any constant over the T -segment vanishes by cuspidality: $\int_{\gamma_{\tau_0}^T} f(\tau) d\tau = c_f(0) = 0$ (Lemma 3.1(c)). Hence the constant difference contributes zero and (26) follows. $\square$

Proposition 3.6 (Integral representation of r_f at the interior monomial). *Let $f \in S_k^!$ and $\tau_0 \in \mathbb{H}$. Let $P \in V_n/\mathbb{C}Y^n$ be the unique solution of $\delta_T P = (2\pi i)^{-(n+1)} C_T^f(\tau_0)$ from Lemma 3.1 and Lemma 3.2, and define the modified-cocycle S -value*

$$r_f := C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P) \in V_n \otimes \mathbb{C}. \quad (27)$$

For each interior $\ell \in \{2, \dots, k-2\}$, the coefficient of the monomial $X^{n-(\ell-1)} Y^{\ell-1}$ in r_f admits the integral representation

$$[r_f]_{X^{n-(\ell-1)} Y^{\ell-1}} = (2\pi i)^{n+1} (-1)^{\ell-1} \binom{n}{\ell-1} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right], \quad (28)$$

with $\tilde{k}_T(\tau, s)$ the Hurwitz-zeta kernel of (3).

Proof. S-piece. The moment expansion (14) gives

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell I_S^\ell(f, \tau_0) X^{n-\ell} Y^\ell,$$

so $[C_S^f(\tau_0)]_{X^{n-(\ell-1)} Y^{\ell-1}} = (2\pi i)^{n+1} \binom{n}{\ell-1} (-1)^{\ell-1} I_S^{\ell-1}(f, \tau_0)$, with $I_S^{\ell-1} = \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau$; this contributes the S -segment piece of (28) against the bare Mellin kernel $\tau^{\ell-1}$.

T-piece, Bernoulli form. S acts on V_n as $(X, Y) \mapsto (-Y, X)$, so for $P = \sum_l p_l X^{n-l} Y^l$ one has $[P|_S - P]_{X^{n-(\ell-1)} Y^{\ell-1}} = (-1)^{\ell-1} p_{n-\ell+1} - p_{\ell-1}$ (n even). By Lemma 3.2, each p_l ($0 \leq l \leq n-1$) is the explicit $\mathbb{Q}$ -linear combination $p_l = \sum_{r=1}^{l+1} \beta_{l,r} I_T^r(f, \tau_0)$ with $\beta_{l,r}$ from (22) of Definition 3.3, obtained by substituting the rescaled T -moments $q_r = \binom{n}{r} (-1)^r I_T^r$ of $(2\pi i)^{-(n+1)} C_T^f(\tau_0)$ into (19). In terms of $\hat{k}_T(\ell, \tau)$ of (23), the T -segment piece of $[r_f]_{X^{n-(\ell-1)} Y^{\ell-1}}$ reads $(2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \hat{k}_T(\ell, \tau) d\tau$.

T-piece, Hurwitz form. By Lemma 3.5, (26), the Bernoulli T -integral of the preceding paragraph equals $(-1)^{\ell-1} \binom{n}{\ell-1} \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau$. Combined with the S -piece, this yields (28) with the bare Hurwitz kernel $\tilde{k}_T(\tau, \ell)$ inside. $\square$

Lemma 3.7 (Brown-basis extraction of $\omega^\pm(f)$). *Let $r_f \in V_n \otimes \mathbb{C}$ be the modified-cocycle S -value of (27). The Eichler–Shimura / Brown cuspidal-basis decomposition ([1, 12]) reads*

$$r_f = \omega^+(f) p_f^+ + \omega^-(f) p_f^- + (\text{Eisenstein-direction term}), \quad (29)$$

with the Eisenstein-direction term contributing only to the boundary monomials X^n, Y^n of r_f (as it is a multiple of the coboundary $Y^n|_S - Y^n = X^n - Y^n$; see Appendix B for the distinction between this

Eisenstein-direction contribution to r_f and the period polynomial of an actual Eisenstein series). Combined with the integral representation (28) of Proposition 3.6, this yields

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right] \quad (30)$$

for every interior $\ell \in \{2, \dots, k-2\}$, with parity $\pm$ determined by $\ell-1 \bmod 2$ and Brown basis coefficient $c_S^{\ell-1} := [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}} \in \mathbb{Q}$. Identity (30) is identity (4) of Theorem 1.1.

Proof. For interior $\ell-1 \in \{1, \dots, n-1\}$, the Eisenstein-direction term in (29) contributes zero at the monomial $X^{n-(\ell-1)}Y^{\ell-1}$, so $[r_f]_{X^{n-(\ell-1)}Y^{\ell-1}} = \omega^\pm(f) c_S^{\ell-1}$. Substituting (28) and dividing through by $c_S^{\ell-1}$ yields (30). $\square$

Corollary 3.8 (Basepoint independence). *The two-segment linear functional of Theorem 1.1, $\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} [\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau]$ of (4), is independent of $\tau_0 \in \mathbb{H}$, and returns the same $\omega^\pm(f)$ for every interior $\ell \in \{2, \dots, k-2\}$ in a fixed parity sector.*

Proof. τ_0 -independence. By (27), namely $r_f := C_S^f(\tau_0) - (2\pi i)^{n+1}(P|_S - P)$, r_f depends on τ_0 only through the cohomology class $[C^f] \in H^1(\Gamma; V_n \otimes \mathbb{C})$ and the choice of coboundary representative P . The class is τ_0 -invariant by the basepoint-shift coboundary identity $C_\gamma^f(\tau'_0) - C_\gamma^f(\tau_0) = P|_\gamma - P$ for the antiderivative $P = (2\pi i)^{n+1}(F(\tau'_0) - F(\tau_0))$ of (16) (valid for f without interior poles), and the coboundary representative $P(\tau_0)$ from Lemma 3.2 is the unique-modulo- $\mathbb{C}Y^n$ solution of $\delta_T P = (2\pi i)^{-(n+1)} C_T^f$. The residual $\mathbb{C}Y^n$ ambiguity lies in the Eisenstein direction: $Y^n|_S - Y^n = X^n - Y^n$ is a coboundary that contributes only to the boundary monomials X^n, Y^n of r_f . Therefore the cuspidal coordinates of r_f — the $\omega^\pm(f)$ — are τ_0 -invariant.

ℓ -independence. This carries no content beyond Lemma 3.7, where r_f is shown to lie in the 1-dimensional cuspidal line $\mathbb{C}p_f^\pm$: reading any interior monomial of r_f and dividing by $c_S^{\ell-1} = [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}}$ returns the same $\omega^\pm(f)$, for every interior ℓ in the parity sector. $\square$

With Proposition 3.6 (integral representation of $[r_f]$ at interior monomials), Lemma 3.4 (integer- s collapse of $\tilde{k}_T$), Lemma 3.7 (Brown-basis extraction of $\omega^\pm$), and Corollary 3.8 (ℓ - and τ_0 -independence) in hand, we assemble the proof of Theorem 1.1. For reference, the theorem statement is:

Theorem 3.9 (Periods and quasi-periods as an explicit linear functional, $\dim S_k \leq 1$ – Theorem 1.1, restated). *Let $k \in 2\mathbb{Z}$ with $\dim S_k \leq 1$, and let $f \in S_k^!$. For the substantive cases $k \in \{12, 16, 18, 20, 22, 26\}$ where $\dim S_k = 1$, the period and quasi-period coefficients $\omega^\pm(f)$ are given by the explicit two-segment linear functional*

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right] \quad (31)$$

for every $f \in S_k^!$, every $\tau_0 \in \mathbb{H}$, and every interior $\ell \in \{2, \dots, k-2\}$, with parity $\pm$ determined by $\ell-1 \bmod 2$. Here $c_S^{\ell-1} := [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}} \in \mathbb{Q}$ is the Brown cuspidal-basis coefficient at the matched monomial ([1, 12]), $\tilde{k}_T(\tau, s)$ is the Hurwitz-zeta T -kernel of (3), and its integer- $s = \ell$ specialisation collapses via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ ($m \geq 0$) to the $\mathbb{Q}$ -rational polynomial

$$\tilde{k}_T(\tau, \ell) = -\frac{B_\ell(\tau+1)}{\ell} + (-1)^{\ell-1} \frac{B_{k-\ell}(\tau+1)}{k-\ell} \in \mathbb{Q}[\tau] \quad (32)$$

of degree at most $k-2$. The right side of (31) is independent of $\tau_0 \in \mathbb{H}$.

Proof of Theorem 1.1, aka Theorem 3.9. The proof proceeds in seven steps. Steps 1–3 motivate the operator δ_T as the natural object whose inversion delivers the cuspidal cocycle representative. Step 4 (Lemma 3.2) performs the inversion in closed form. Steps 5–7 read $\omega^\pm(f)$ off the resulting period polynomial.

Step 1: the canonical cuspidal representative has $C_T^f = 0$. Within the cohomology class $[C^f] \in H^1(\Gamma; V_n)$, consider the representative with vanishing T -component. When $C_T = 0$, the cocycle relations $C_{S^2} = 0$ and $C_{(ST)^3} = 0$ collapse onto C_S alone, and the surviving relations on C_S are exactly the two Eichler–Shimura period-polynomial relations $r_f|(1+S) = 0$ and $r_f|(1+U+U^2) = 0$ with $U = TS$. The $C_T = 0$ representative therefore has its S -value in the Eichler–Shimura period-polynomial space, where Brown’s basis $\{p_f^+, p_f^-, \text{Eisenstein}\}$ is defined and the cuspidal coordinates are $\omega^\pm(f)$. The proof below kills C_T^f by a coboundary modification.

Step 2: kill C_T^f by a coboundary modification. The only freedom inside the cohomology class is to add a coboundary $\delta P(\gamma) := P|_\gamma - P$ for some $P \in V_n$. Under this modification, $C_T^f(\tau_0) \mapsto C_T^f(\tau_0) - (P|_T - P)$. Killing C_T^f becomes the equation

$$\delta_T P := P|_T - P = C_T^f(\tau_0), \quad (33)$$

to be solved for $P \in V_n$. This identifies the linear endomorphism $\delta_T: V_n \rightarrow V_n$ of Lemma 3.1 as the precise object whose inversion delivers the desired coboundary representative.

Step 3: solvability of $\delta_T P = C_T^f$ is exactly cuspidality of f . Lemma 3.1 computes $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} c_f(0)$ and shows that $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$. Solvability of (33) is therefore equivalent to $c_f(0) = 0$, i.e. to f being a (weak) cusp form. The cuspidality hypothesis $f \in S_k^!$ is not a separate input to the construction: it is the solvability condition of the coboundary equation in Step 2.

Step 4: closed-form Bernoulli inversion of δ_T . Lemma 3.2 inverts δ_T in closed form, giving the coboundary representative $P = \delta_T^{-1}((2\pi i)^{-(n+1)} C_T^f(\tau_0)) \in V_n/\mathbb{C}Y^n$ explicitly in Bernoulli numbers.

Step 5: the modified cocycle and its S -component r_f . With $P := \delta_T^{-1}((2\pi i)^{-(n+1)} C_T^f(\tau_0)) \in V_n/\mathbb{C}Y^n$ from Step 4, the modified cocycle $C'^f := C^f - (2\pi i)^{n+1} \delta P$ has $C_T'^f = 0$ by construction, and its S -value $r_f := C_S'^f(\tau_0) = C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P)$ of (27) is the Eichler–Shimura period polynomial of f . By the Eichler–Shimura / Brown cuspidal-basis decomposition (29) of Lemma 3.7, $\omega^\pm(f)$ are the explicit linear coordinates of r_f along $p_f^\pm$ in the 1-dimensional cuspidal direction.

Step 6: unpacking r_f into bare Mellin and Hurwitz integrals. Substituting the moment expansion of $C_S^f(\tau_0)$ from (14) and the Bernoulli expression for P from Step 4 into (27), and reading off the coefficient of $X^{n-(\ell-1)} Y^{\ell-1}$ for each interior $\ell \in \{2, \dots, k-2\}$, Proposition 3.6 produces the integral representation (28) of $[r_f]_{X^{n-(\ell-1)} Y^{\ell-1}}$ with the bare Mellin kernel $\tau^{\ell-1}$ on the S -segment and the bare Hurwitz kernel $\tilde{k}_T(\tau, \ell)$ on the T -segment. Lemma 3.4 (the integer- s collapse of $\tilde{k}_T$) exhibits the T -kernel as a $\mathbb{Q}$ -rational polynomial of degree $\leq k-2$. Lemma 3.7 (Brown-basis extraction) then divides through by $c_S^{\ell-1}$ to give (4).

Step 7: ℓ - and τ_0 -independence. Corollary 3.8 supplies both: r_f is a single polynomial in the 1-dimensional cuspidal direction, so every interior monomial returns the same $\omega^\pm(f)$ (parity-selected by ℓ), while τ_0 -invariance follows from $[C^f]$ being a cohomology-class invariant (the residual $\mathbb{C}Y^n$ ambiguity touching only the Eisenstein direction). $\square$

3.1 Numerical verification

At $k = 12$, every step of the period-extraction pipeline of Theorem 1.1 has been verified against Brown's published values [1] for the canonical pair $f \in \{\Delta, \hat{\Delta}\}$ in the companion script `verify_periods.py` [4], run at 50-digit `mpmath` precision.

	$\Delta \in S_{12}$, period $\omega^\pm(\Delta)$	$\hat{\Delta} \in S_{12}^!$, quasi-period $\eta^\pm(\hat{\Delta})$
+	-68 916 772.80959519...	127 202 100 647.17709...
$-/i$	- 5 585 015.37931040...	10 276 732 343.64913...

Table 1: Periods of Δ and quasi-periods of $\hat{\Delta}$ at $k = 12$, reproduced from the Fourier coefficients of f by (4) to relative residual $\lesssim 10^{-42}$ at $\tau_0 = 0.3 + 1.2i$. Reference values from [1].

The verification is layered through the proof of Theorem 1.1 as follows; the relevant named functions in `verify_periods.py` [4] are indicated in `typewriter`.

- **Cocycle integration** ((12), (14)): `cocycle_poly` computes $C_S^f(\tau_0), C_T^f(\tau_0) \in V_{10} \otimes \mathbb{C}$ by V_{10} -valued contour integration along $\gamma_{\tau_0}^{S,T}$.
- **Coboundary inversion two ways** (Lemma 3.2): `solve_coboundary_LU` solves $\delta_T P = (2\pi i)^{-(n+1)} C_T^f$ by forward substitution on the strictly-lower-triangular matrix (18); `solve_coboundary_bernoulli` uses the closed form (19) of Lemma 3.2. The two outputs agree to relative residual $\sim 10^{-40}$, confirming (19).
- **Brown-basis extraction at 9 interior monomials** (Lemma 3.7): `extract_periods` reads off $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ from $r_f = C_S^f - (2\pi i)^{n+1}(P|_S - P)$ at each interior $\ell \in \{2, \dots, 10\}$. That all 9 agree to $\lesssim 10^{-42}$, matching the Brown values of Table 1, is the numerical check that the computed r_f lands on the cuspidal line (Corollary 3.8).
- **Basepoint independence** (Corollary 3.8): re-extraction of $\eta^\pm(\hat{\Delta})$ at $\tau'_0 = -0.2 + 0.9i$ agrees with the $\tau_0 = 0.3 + 1.2i$ values to $\lesssim 10^{-42}$.
- **Clean-form identity** (4) (Theorem 1.1): `hurwitz_extract` evaluates the bare-Mellin + bare-Hurwitz integral form directly at $\ell \in \{2, 3, 4, 5\}$ for $f = \Delta$, agreeing with the Brown reference values in Table 1 to $\lesssim 10^{-42}$.
- **S - vs T -segment growth-rate cancellation** for $f = \hat{\Delta}$: `bare_integrals` confirms that $|I_S|, |I_T|$ each grow $\sim 10^7$ -fold across $\text{Im}(\tau_0) \in [1.2, 2.5]$ while $I_S + I_T$ stays τ_0 -invariant.

Analogous verification at higher weights $k \in \{16, 18, 20, 22, 26\}$ where $\dim S_k = 1$ requires only the Brown basis coefficients $c_S^{\ell-1}$ at those weights and is structurally identical.

4 The Hurwitz-zeta extension and the L -function on $M_k^!$: proof of Theorem 1.2

4.1 Overview and the origin of the kernel

We prove Theorem 1.2: the functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ of (6), built from the Hurwitz-zeta T -kernel $\tilde{k}_T(\tau, s)$ of (3), is finite and basepoint-independent on $M_k^!$, and at the elliptic basepoint $\tau_0 = i$ collapses to the completed L -function in closed form. The argument has three movements: (i) the kernel obeys a shift-difference identity (Lemma 4.1) that renders the two-segment functional finite (Lemma 4.2) and locally constant in its contour data (Lemma 4.3), so $L^*(f, s)$ is well defined; (ii) at $\tau_0 = i$ the S -segment degenerates and a single per-mode *unfolding* (Proposition 4.4) turns the surviving T -integral into incomplete gammas; (iii) the resulting sum is the completed L -function. Lemmas 4.2 and 4.3 are stated for arbitrary meromorphic f , in preparation for §5; Theorem 1.2 is their specialisation to $f \in M_k^!$, where no interior pole obstructs the contour pair.

Why a Hurwitz-zeta kernel. The kernel (3) is the all- s continuation of the integer- s inversion of §3. There, killing the T -cocycle meant solving the shift-difference equation $\delta_T P = P|_T - P = Q$ on V_n , and the operator identity $\delta_T = e^{Y\partial_x} - 1$ inverted it through the Bernoulli generating function $z/(e^z - 1) = \sum_k B_k z^k/k!$ (Lemma 3.2). Dropping the polynomial restriction, the kernel must solve the same shift-difference equation with a power-law right-hand side — the identity (34) of Lemma 4.1 below, whose right side is a combination of pure powers $\tau^{\sigma-1}$. The function that inverts the unit shift against a single power is the Hurwitz zeta, $\zeta(\sigma, \tau + 1) - \zeta(\sigma, \tau) = -\tau^{-\sigma}$, so a combination of Hurwitz zetas is the natural solution, and the Bernoulli inversion of §3 is exactly its integer- s shadow through $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ — the Bernoulli numbers being zeta values at non-positive integers. An alternative derivation, starting from the classical Mellin transform and demanding contour-invariance, is recorded in Appendix A; the present section is self-contained without it.

4.2 Well-definedness of the two-segment functional

Write $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ for the upper half-plane with the Γ -orbit of the interior poles of f removed, and call a contour pair $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ *admissible* if $\tau_0 \in \mathbb{H}_f$ and both segments lie in $\mathbb{H}_f$. For $f \in M_k^!$ there are no interior poles, so $\mathbb{H}_f = \mathbb{H}$ and every pair is admissible.

Lemma 4.1 (Shift-difference identity for $\tilde{k}_T$). *The kernel $\tilde{k}_T(\tau, s)$ of (3) is analytic in $\tau \in \mathbb{H}$ and meromorphic in $s \in \mathbb{C}$, and satisfies*

$$\tilde{k}_T(\tau, s) - \tilde{k}_T(\tau - 1, s) = -\tau^{s-1} + e^{i\pi(s-1)} \tau^{k-1-s} \quad (34)$$

for every $\tau \in \mathbb{H}$ and every $s \in \mathbb{C}$.

Proof. Analyticity in τ is immediate: each Hurwitz factor $\zeta(\cdot, \tau + 1)$ has second argument with $\text{Re}(\tau + 1) > 1$ on $\mathbb{H}$, where $\zeta(\sigma, a)$ is analytic in a . For the identity, the Hurwitz raising relation $\zeta(\sigma, a + 1) - \zeta(\sigma, a) = -a^{-\sigma}$ [6, §25.11.3] at $(\sigma, a) = (1 - s, \tau)$ and at $(\sigma, a) = (s - (k - 1), \tau)$ gives

$$\zeta(1 - s, \tau + 1) - \zeta(1 - s, \tau) = -\tau^{s-1}, \quad \zeta(s - (k - 1), \tau + 1) - \zeta(s - (k - 1), \tau) = -\tau^{k-1-s}.$$

Since $\tilde{k}_T(\tau, s) = \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(s - (k - 1), \tau + 1)$ and, shifting $\tau \mapsto \tau - 1$, $\tilde{k}_T(\tau - 1, s) = \zeta(1 - s, \tau) - e^{i\pi(s-1)} \zeta(s - (k - 1), \tau)$, subtracting gives $\tilde{k}_T(\tau, s) - \tilde{k}_T(\tau - 1, s) = -\tau^{s-1} + e^{i\pi(s-1)} \tau^{k-1-s}$, which is (34). $\square$

Lemma 4.2 (Finiteness). *Let f be any meromorphic modular form of weight k on Γ . For every admissible contour pair $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and every $s \in \mathbb{C}$, both integrals in (6) are finite, so $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is well defined. In particular this holds for every $f \in M_k^!$ and every $\tau_0 \in \mathbb{H}$.*

Proof. By admissibility both segments are compact and lie in $\mathbb{H}_f$, on which f is holomorphic; the kernels τ^{s-1} and $\tilde{k}_T(\tau, s)$ are analytic in $\tau \in \mathbb{H}$ for each fixed s (Lemma 4.1). A continuous integrand on a compact contour has finite integral. For $f \in M_k^!$ there are no interior poles, $\mathbb{H}_f = \mathbb{H}$, and every $\tau_0 \in \mathbb{H}$ yields an admissible pair. $\square$

Lemma 4.3 (Local constancy and winding-invariance). *Let f be any meromorphic modular form of weight k on Γ . Then $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is unchanged under any continuous deformation of the admissible data $(\tau_0, \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ that does not move a segment across a point of $\Gamma \cdot f^{-1}(\infty)$; equivalently, it depends on the admissible pair only through the winding of the two segments about the interior poles of f . In particular, for $f \in M_k^!$ — where $\mathbb{H}_f = \mathbb{H}$ is simply connected — $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of $\tau_0 \in \mathbb{H}$ and of the contour pair for every $s \in \mathbb{C}$; we write $L^*(f, s)$ for the common value.*

Proof. Two kinds of variation can change the data: deforming the path shapes with endpoints fixed, and moving the basepoint τ_0 (which drags the endpoints $-1/\tau_0, \tau_0 - 1, \tau_0$).

Path shape. If a segment is deformed within $\mathbb{H}_f$ with its endpoints fixed and without crossing a pole, the integrand $f(\tau) K(\tau, s)$ (with $K \in \{\tau^{s-1}, \tilde{k}_T(\tau, s)\}$) is holomorphic on the swept region, so the integral is unchanged by Cauchy's theorem.

Basepoint. Differentiating (6) in τ_0 — legitimate by the finiteness of Lemma 4.2 — only the endpoints contribute. The T -segment endpoints give $f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s)$, and the S -segment endpoints give $f(\tau_0) \tau_0^{s-1} - f(-1/\tau_0) (-1/\tau_0)^{s-1} \tau_0^{-2}$. Using the modularity relations $f(\tau_0 - 1) = f(\tau_0)$ and $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ together with $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{1-s}$, the total τ_0 -derivative of L^* reduces to

$$i^{-s} f(\tau_0) \left[\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) + \tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s} \right] = 0$$

by the shift-difference identity (34) of Lemma 4.1. Hence L^* is constant under any admissible motion of τ_0 .

Combining the two, L^* is locally constant on the space of admissible configurations and can change only when a segment crosses a pole: it depends only on the winding data. For $f \in M_k^!$ there are no interior poles, $\mathbb{H}_f = \mathbb{H}$ is simply connected, every admissible configuration is joined to every other without crossing a pole, and $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is therefore globally independent of τ_0 and of the contour pair. $\square$

4.3 The unfolding engine

Proposition 4.4 (Hurwitz unfolding). *For every nonzero integer n and every $\nu \in \mathbb{C}$,*

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n), \quad (35)$$

where for $n < 0$ the right side is read on the analytic continuation with $\arg(2\pi n) = +\pi$. At the integer values $\nu \in \{1-s, s-(k-1)\}$ used below (with s an integer in the critical strip), $1-\nu$ is a positive integer N and $\Gamma(1-\nu, x)$ is the entire polynomial-times-exponential $\Gamma(N, x) = (N-1)! e^{-x} \sum_{j=0}^{N-1} x^j / j!$, single-valued for either sign of n .

Proof. For $\operatorname{Re}(\nu) > 1$ the series $\zeta(\nu, \sigma) = \sum_{m \geq 0} (\sigma + m)^{-\nu}$ converges uniformly on $[i, i+1]$ [6, §25.11]; substituting $u = \sigma + m$, the periodicity $e^{2\pi i n m} = 1$ glues the unit segments into a single ray — the *unfolding*, after Rankin–Selberg:

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = \int_i^{i+\infty} e^{2\pi i n u} u^{-\nu} du. \quad (36)$$

Substituting $t = -2\pi i n u$ and recalling $\Gamma(\sigma, x) = \int_x^\infty t^{\sigma-1} e^{-t} dt$ [6, §8.2.2] evaluates the ray integral as

$$\int_i^{i+\infty} e^{2\pi i n u} u^{-\nu} du = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n), \quad (37)$$

which is (35) for $\operatorname{Re}(\nu) > 1$ (for $n < 0$ the path runs to $+i\infty$, giving the $\arg(2\pi n) = +\pi$ continuation of Γ). Both sides are entire in ν (Γ by [6, §8.2.7]; the left integral since the ζ -pole at argument 1 integrates to zero over a unit segment), so the identity extends to all $\nu \in \mathbb{C}$. $\square$

The same unfolding at an arbitrary base point $\tau_0 \in \mathbb{H}$ in place of i — the translate-gluing $\sum_{m \geq 0} \int_{\tau_0+m}^{\tau_0+m+1} = \int_{\tau_0}^{\tau_0+\infty}$ is base-point-independent — gives, for every nonzero integer n and every $\nu \in \mathbb{C}$,

$$\int_{\tau_0}^{\tau_0+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n \tau_0), \quad (38)$$

whose second argument has $\operatorname{Re}(-2\pi i n \tau_0) = 2\pi n \operatorname{Im}(\tau_0)$ — positive for $n > 0$, negative for $n < 0$ — with the $\arg = +\pi$ continuation in the latter case. Equation (35) is the case $\tau_0 = i$; we use (38) in §5.

Lemma 4.5 (Constant-mode integral). *For every $s \in \mathbb{C}$,*

$$\int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -\frac{i^s}{s} + e^{i\pi(s-1)} \frac{i^{k-s}}{k-s}. \quad (39)$$

Proof. The Hurwitz zeta has antiderivative $\int \zeta(\sigma, a) da = -\zeta(\sigma-1, a)/(\sigma-1) + C$ for $\sigma \neq 1$; evaluating across the shifted segment and applying the raising identity $\zeta(\sigma-1, a+1) - \zeta(\sigma-1, a) = -a^{1-\sigma}$ at $a = i$ gives

$$\int_{i-1}^i \zeta(\sigma, \tau+1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma-1}. \quad (40)$$

Applying (40) to the two halves of $\tilde{k}_T(\tau, s)$ at $\sigma = 1-s$ and $\sigma = s-(k-1)$ yields (39). $\square$

4.4 Restated theorem and proof

For reference, the theorem statement is the following.

Theorem 4.6 (*L*-function for weakly holomorphic forms – Theorem 1.2, restated). *Let $k \in 2\mathbb{Z}$ and $f \in M_k^1$. With $\tilde{k}_T(\tau, s)$ the Hurwitz-zeta kernel of (3), for every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right] \quad (41)$$

is finite and independent of both $\tau_0 \in \mathbb{H}$ and of the admissible contour choices $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$. We write $L^(f, s)$ for the common value.*

Specialising to the elliptic fixed point $\tau_0 = i$ of the modular involution $S: \tau \mapsto -1/\tau$, the S -segment $[-1/i, i]$ collapses to the single point i , the S -integral in (41) vanishes, and L^* reduces to a single integral on the compact contour $[i-1, i] \subset \mathbb{H}$, with closed-form sum

$$L^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right]. \quad (42)$$

The right side of (42) is the completed L -function $L^*(f, s)$ of f . For holomorphic modular forms it reduces to the classical Hecke–Weil Mellin transform. For weak modular forms, it reduces to the results of [2], proving that their non-holomorphic regulator prescription gives the unique, canonical L -function for every weak modular form.

Proof of Theorem 1.2, aka Theorem 4.6. The proof runs in seven steps: well-definedness (Steps 1–2), the collapse at $\tau_0 = i$ (Step 3), the per-mode evaluation (Steps 4–6), and the identification (Step 7).

Step 1: finiteness. For $f \in M_k^!$ one has $\mathbb{H}_f = \mathbb{H}$, so every $\tau_0 \in \mathbb{H}$ gives an admissible contour pair and both integrals in (41) are finite for every $s \in \mathbb{C}$ by Lemma 4.2.

Step 2: τ_0 - and contour-independence. By Lemma 4.3, specialised to the simply-connected $\mathbb{H}_f = \mathbb{H}$ of a weak form, $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of τ_0 and of the admissible contour pair. This establishes the well-definedness assertion of (41) and licenses writing $L^*(f, s)$ for the common value.

Step 3: collapse at $\tau_0 = i$. At the elliptic fixed point $\tau_0 = i$ the S -segment endpoints coincide, $-1/i = i$, so γ_i^S has zero length and the S -integral in (41) vanishes. By Step 2 the value is unchanged, leaving

$$L^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau. \quad (43)$$

Step 4: per-mode expansion. Expanding $f = \sum_n c_f(n) q^n$ in (43) and swapping sum and integral — justified by uniform convergence of the q -series on the compact segment $[i-1, i]$, including the finitely many polar modes, since $|q^{-1}| = e^{2\pi}$ is bounded there — gives

$$L^*(f, s) = i^{-s} \sum_n c_f(n) \int_{i-1}^i e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau. \quad (44)$$

Step 5: the constant mode. For $n = 0$, Lemma 4.5 and the phase collapse $i^{-s} e^{i\pi(s-1)} i^{k-s} = -i^k$ give the contribution

$$c_f(0) \cdot i^{-s} \int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right), \quad (45)$$

the polar term of [3, Lemma 2.1] (equivalently the polar piece of [2, Thm. 1.1]) at the splitting point $B = 1$.

Step 6: the non-constant modes. For $n \neq 0$, substituting $\sigma = \tau + 1$ shifts the contour to $[i, i + 1]$ with prefactor $e^{-2\pi in} = 1$, and Proposition 4.4 applied to the two halves of $\tilde{k}_T(\tau, s) = \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)}\zeta(s - (k - 1), \tau + 1)$ at $\nu = 1 - s$ and $\nu = s - (k - 1)$ gives

$$\int_i^{i+1} e^{2\pi in\sigma} \zeta(1 - s, \sigma) d\sigma = i^s (2\pi n)^{-s} \Gamma(s, 2\pi n), \quad (46)$$

$$\int_i^{i+1} e^{2\pi in\sigma} \zeta(s - (k - 1), \sigma) d\sigma = i^{k-s} (2\pi n)^{s-k} \Gamma(k - s, 2\pi n). \quad (47)$$

The first half against the i^{-s} prefactor gives $\Gamma(s, 2\pi n)/(2\pi n)^s$; the second against the relative phase $-e^{i\pi(s-1)}$ collapses via $-i^{-s} e^{i\pi(s-1)} i^{k-s} = i^k$ to $i^k \Gamma(k - s, 2\pi n)/(2\pi n)^{k-s}$. Hence the per-mode summand is

$$c_f(n) \cdot i^{-s} \int_{i-1}^i e^{2\pi in\tau} \tilde{k}_T(\tau, s) d\tau = c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k - s, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (48)$$

the $B = 1$ value of the cuspidal summand of [3, Lemma 2.1]. Summing (45) and (48) over all Fourier modes in (44) reproduces the closed-form sum (42).

Step 7: identification. The right side of (42) is the completed L -function $L^*(f, s)$: for holomorphic f it is the classical Hecke–Weil Mellin transform, and for weak f it is the incomplete-gamma sum of [2], here obtained at every $s \in \mathbb{C}$ from a finite contour with no regulator. This completes the proof. $\square$

4.5 Worked example at $k = 12$: Bernoulli polynomials and numerics

At $k = 12$ the kernel of (3) is

$$\tilde{k}_T(\tau, s) = \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(s - 11, \tau + 1).$$

At integer values of $s \in \{1, 2, \dots, 11\}$, both Hurwitz zeta factors collapse to Bernoulli polynomials via $\zeta(-m, a) = -B_{m+1}(a)/(m + 1)$ ($m \geq 0$), and $\tilde{k}_T(\tau, s)$ becomes a polynomial of degree ≤ 11 in τ . Three representative values:

$$\tilde{k}_T(\tau, 1) = -\tau - \frac{1}{2} + \frac{1}{11} B_{11}(\tau + 1), \quad \tilde{k}_T(\tau, 6) = -\frac{1}{3} B_6(\tau + 1),$$

$$\tilde{k}_T(\tau, 11) = -\frac{1}{11} B_{11}(\tau + 1) + \tau + \frac{1}{2}.$$

The $s \leftrightarrow 12 - s$ symmetry follows from the $s \leftrightarrow k - s$ structure of the two-term kernel (with $i^k = +1$ at $k = 12$).

Identification with Theorem 1.1. With the clean form of Theorem 1.1, the T -segment kernel of (4) is exactly the integer- $s = \ell$ specialisation $\tilde{k}_T(\tau, \ell)$ of the Hurwitz-zeta kernel of (3), with no convention conversion required: all combinatorial structure $(-1)^{\ell-1} \binom{k-2}{\ell-1} / c_S^{\ell-1}$ governing the $\omega^\pm(f)$ normalisation lives outside the two-segment bracket. The period-extraction functional (4) is thus the rescaled integer- s shadow of the all- s L -function functional (6) of Theorem 1.2, with the bare Mellin τ^{s-1} and Hurwitz $\tilde{k}_T(\tau, s)$ kernels mapped to their integer- $s = \ell$ specialisations and the analytic prefactor i^{-s} replaced by the combinatorial Brown-basis factor.

Numerical values. The companion `mpmath` scripts [4], run at 50-digit precision with $f \in \{\Delta, \widehat{\Delta}\}$, reproduce the periods $\omega^\pm(\Delta)$ and quasi-periods $\eta^\pm(\widehat{\Delta})$ of [1] listed in Table 1, to relative residual $\lesssim 10^{-40}$, the same for every interior $\ell \in \{2, \dots, 10\}$ and every basepoint $\tau_0 \in \{0.3+1.2i, i, 1.5i, -0.4+0.8i\}$.

Complex- s verification. Beyond the integer- ℓ period extraction, the all- s functional (6) was checked directly against the canonical completed L -function $L^*(\Delta, s)$ of Theorem 1.2 at non-integer s [4]. At half-integer points $s \in \{1.5, 2.5, \dots, 6.5\}$, off-axis complex points $s \in \{2.7 + 0.4i, 5 + i, 5.5 + 2.5i\}$, and points outside the critical strip $s \in \{0.5, -0.3\}$, the finite-contour value agrees with $L^*(\Delta, s)$ to relative residual $\sim 10^{-53}$ at 50-digit precision, and is independent of $\tau_0 \in \{0.3 + 1.2i, 1.5i, -0.4+0.8i, 0.2+2i\}$ at the same level. The corresponding check for $\widehat{\Delta}$ holds to $\sim 10^{-47}$, the loss of digits reflecting the q^{-1} cusp mode.

5 Wall-crossing for meromorphic forms: proof of Theorem 1.3

5.1 Overview

Theorem 1.3 has three parts. Parts (a) (finiteness) and (b) (dependence on the admissible contour pair only through its homotopy class in $\mathbb{H}_f$) hold for arbitrary meromorphic f and were already proved, without any cuspidality hypothesis, as Lemma 4.2 and Lemma 4.3 of §4 — stated there in exactly this generality so as to serve here. The new content of this section is part (c): the explicit residue $\mathcal{R}_{\tau_p}(f, s)$ by which L^* jumps when the contour pair crosses an interior pole (Lemma 5.1, Proposition 5.2), assembled into the proof of Theorem 1.3 below. We then reduce L^* to the deformed-Mellin L -function $L_f^*(s)_{\text{int}}$ of [3] on forms regular at the cusp (Corollary 1.4), record the integer- s period-polynomial form of the wall-crossing (Corollary 5.6), and state the per-mode closed form for interior-pole forms regular at the cusp (Proposition 5.7).

5.2 The wall-crossing residue

We compute the jump in L^* when a contour deformation drags one segment across a single interior pole.

Lemma 5.1 (Single-segment crossing). *Let f be a meromorphic modular form of weight k with a simple pole at $\tau_p \in \mathbb{H}$, and let $K(\tau, s) \in \{\tau^{s-1}, \widetilde{k}_T(\tau, s)\}$ be either kernel. Suppose a continuous deformation carries a segment from Σ_0 to Σ_1 (within $\mathbb{H}_f$ away from τ_p) so that the closed loop $\Sigma_1 - \Sigma_0$ encircles τ_p once with orientation $\sigma \in \{+1, -1\}$ and no other pole. Then*

$$\int_{\Sigma_1} f(\tau) K(\tau, s) d\tau - \int_{\Sigma_0} f(\tau) K(\tau, s) d\tau = 2\pi i \sigma \text{Res}(f, \tau_p) K(\tau_p, s). \quad (49)$$

Proof. The kernel $K(\cdot, s)$ is analytic on $\mathbb{H}$, so $f(\tau) K(\tau, s)$ has a simple pole at τ_p with residue $\text{Res}(f, \tau_p) K(\tau_p, s)$ and no other singularity in the region swept between Σ_0 and Σ_1 . The difference of the two integrals is the integral over the closed loop $\Sigma_1 - \Sigma_0$, which by the residue theorem equals $2\pi i \sigma \text{Res}(f, \tau_p) K(\tau_p, s)$. $\square$

Proposition 5.2 (Wall-crossing residue). *Let f be a meromorphic modular form of weight k , and let $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and $(\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$ be two admissible contour pairs whose homotopy classes in $\mathbb{H}_f$ differ by*

a single crossing of a simple interior pole $\tau_p \in \mathbb{H}$ of f , with S - and T -segment winding numbers $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$. Then

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) - L^*(f, s; \gamma_{\tau_0'}^S, \gamma_{\tau_0'}^T) = -2\pi i i^{-s} \operatorname{Res}(f, \tau_p) \left[\sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (50)$$

which is the residue $\mathcal{R}_{\tau_p}(f, s)$ of (10).

Proof. Apply Lemma 5.1 to the S -segment with kernel τ^{s-1} and winding $\sigma_{\tau_p}^S$, and to the T -segment with kernel $\tilde{k}_T(\tau, s)$ and winding $\sigma_{\tau_p}^T$. Carried through the common prefactor i^{-s} of (2), the two contributions sum to (50). At a generic wall exactly one of $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T$ is nonzero, according to which segment crosses; both are nonzero only on the codimension-two locus where the two segments meet Γ -images of τ_p simultaneously. $\square$

For a pole of order $m \geq 1$ the same computation applies verbatim, with $\operatorname{Res}(f, \tau_p) K(\tau_p, s)$ in (49) replaced by the order- m residue $\sum_{\ell=1}^m r_{f, \tau_p}(-\ell) \partial_{\tau_p}^{\ell-1} K(\tau_p, s) / (\ell-1)!$, where $r_{f, \tau_p}(-\ell)$ are the principal-part Laurent coefficients of f at τ_p ; the generalisation is routine, and we record only the simple-pole case.

5.3 Restated theorem and proof

For reference, the theorem statement is the following.

Theorem 5.3 (L -function for meromorphic modular forms – Theorem 1.3, restated). *Let $k \in 2\mathbb{Z}$ and let f be a meromorphic modular form of weight k on Γ . Write $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ for the upper half-plane with the Γ -orbit of the interior poles of f removed, and call the pair $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ admissible if $\tau_0 \in \mathbb{H}_f$ and both contours lie in $\mathbb{H}_f$, with $\gamma_{\tau_0}^S$ running from $S^{-1}\tau_0 = -1/\tau_0$ to τ_0 and $\gamma_{\tau_0}^T$ from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 . Define $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ by (2) with the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ of (3). Then:*

- (a) Both integrals in (2) are finite for every admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and every $s \in \mathbb{C}$.
- (b) $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ depends on the admissible pair only through its homotopy class in $\mathbb{H}_f$: any continuous one-parameter family of admissible pairs — including variations of the anchoring basepoint τ_0 — leaves the value unchanged.
- (c) If two admissible configurations $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ and $(\gamma_{\tau_0'}^S, \gamma_{\tau_0'}^T)$ lie in homotopy classes related by a single crossing of an interior pole $\tau_p \in \mathbb{H}$ of f , then

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) - L^*(f, s; \gamma_{\tau_0'}^S, \gamma_{\tau_0'}^T) = \mathcal{R}_{\tau_p}(f, s), \quad (51)$$

where for a simple pole $\operatorname{Res}(f, \tau_p)$,

$$\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \operatorname{Res}(f, \tau_p) \left[\sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (52)$$

with $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$ the winding-number contributions of $\gamma_{\tau_0}^S$ and $\gamma_{\tau_0}^T$ across τ_p (the generic case has one nonzero and the other zero, according to which contour crossed). Higher-order poles contribute via the corresponding derivatives in τ_p of (52), as recorded after Proposition 5.2.

Proof of Theorem 1.3, aka Theorem 5.3. (a) Finiteness of both integrals for any admissible pair is Lemma 4.2. (b) Dependence on the admissible pair only through its homotopy class in $\mathbb{H}_f$ — invariance under any admissible deformation, the basepoint included — is Lemma 4.3. (c) When two admissible classes differ by a single crossing of a simple interior pole τ_p , the jump (51) is (50) of Proposition 5.2, equal to $\mathcal{R}_{\tau_p}(f, s)$ of (52); the order- m case is the higher residue recorded after Proposition 5.2. $\square$

5.4 Reduction on regular-at-cusps meromorphic forms

The two-segment functional makes the cusp behaviour of the T -segment completely explicit: it is the BFK incomplete-gamma sum with the split height promoted to the endpoint τ_0 . We record this closed form and its asymptotics, then specialise to forms regular at the cusp to reduce L^* to the deformed-Mellin L -function $L_f^*(s)_{\text{int}}$ of [3].

Lemma 5.4 (*T -segment closed form and cusp asymptotics*). *Let $f \in M_k^!$ with Fourier expansion $f = \sum_n c_f(n) q^n$. For every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,*

$$i^{-s} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left[\frac{(\tau_0/i)^s}{s} + i^k \frac{(\tau_0/i)^{k-s}}{k-s} \right] + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right]. \quad (53)$$

Consequently, as $\text{Im}(\tau_0) \rightarrow \infty$ with $\text{Re}(\tau_0)$ fixed:

- (a) *if f is regular at the cusp with $f = O(q^N)$, $N \geq 1$, the T -segment integral decays as $O(e^{-2\pi N \text{Im}(\tau_0)})$ (up to a polynomial in τ_0);*
- (b) *if f has a cusp pole $f = c_f(-M)q^{-M} + \dots$ with $c_f(-M) \neq 0$, $M \geq 1$, the T -segment integral grows as $|c_f(-M)|e^{2\pi M \text{Im}(\tau_0)}$, matching the cusp-pole growth of f itself.*

Proof. Expand $f = \sum_n c_f(n) q^n$ on the compact segment $[\tau_0-1, \tau_0]$ (uniformly convergent — the finitely many polar modes are bounded there, $|q|$ being bounded below) and integrate term by term. For $n \neq 0$, the base- τ_0 unfolding (38) applied to the two halves of $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)}\zeta(s-(k-1), \tau+1)$ at $\nu = 1-s$ and $\nu = s-(k-1)$, with the phase collapse $-i^{-s}e^{i\pi(s-1)}i^{k-s} = i^k$ of §4.4, gives the $n \neq 0$ summand of (53). For $n = 0$, the Hurwitz antiderivative of Lemma 4.5 at base τ_0 , $\int_{\tau_0-1}^{\tau_0} \zeta(\sigma, \tau+1) d\tau = \tau_0^{1-\sigma}/(\sigma-1)$, produces the bracketed constant-mode term (which at $\tau_0 = i$ reduces to the $-c_f(0)(1/s + i^k/(k-s))$ of (7)).

For the asymptotics, the constant-mode term is a polynomial in τ_0 , hence sub-exponential. In each $n \neq 0$ summand the incomplete-gamma argument satisfies $\text{Re}(-2\pi i n \tau_0) = 2\pi n \text{Im}(\tau_0)$, and the uniform asymptotic $\Gamma(\sigma, z) \sim z^{\sigma-1}e^{-z}$ as $|z| \rightarrow \infty$ for $|\arg z| < \frac{3\pi}{2}$ [6, §8.11] gives $|c_f(n)|$ -weighted decay $e^{-2\pi n \text{Im}(\tau_0)}$ for $n > 0$ and growth $e^{2\pi |n| \text{Im}(\tau_0)}$ for $n < 0$. Taking the lowest present mode yields (a) and (b). $\square$

For $f \in F_k$ regular at the cusp ($f = \sum_{n \geq N} c_f(n) q^n$, $N \geq 0$), part (a) of Lemma 5.4 drives the T -segment to zero as $\text{Im} \tau_0 \rightarrow \infty$, which gives the reduction to $L_f^*(s)_{\text{int}}$.

Corollary 5.5 (*L^* on regular-at-cusps meromorphic forms – Corollary 1.4, restated*). *Let $f \in F_k$ be a meromorphic modular form of weight k that is regular at the cusp. Let $L_f^*(s)_{\text{int}, \gamma}$ denote the deformed-Mellin L -function of f along a contour γ from 0 to $i\infty$ lying in $\mathbb{H}_f$ (see [3, §2]). Then for every admissible $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ whose homotopy class in $\mathbb{H}_f$ is compatible with γ (in the sense that the contour $\gamma_{\tau_0}^S$ together with the cusp-limit piece $\lim_{\tau_0 \rightarrow i\infty} \gamma_{\tau_0}^S \rightarrow [0, i\infty]$ can be deformed to γ in $\mathbb{H}_f$),*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_f^*(s)_{\text{int}, \gamma} \quad \text{for every } s \in \mathbb{C}. \quad (54)$$

Proof of Corollary 1.4, aka Corollary 5.5. Take $\tau_0 = \varepsilon + i\Lambda$ with $\varepsilon \in \mathbb{R}$ chosen so that $\text{Re}(\tau) = \varepsilon$ avoids every Γ -image of an interior pole of f , fixing the homotopy class compatible with γ . By Lemma 4.3 the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of Λ within that class. As $\Lambda \rightarrow \infty$, the T -segment integral tends to 0 by Lemma 5.4 (for $N \geq 1$; for $N = 0$ the same holds after the constant-term subtraction of [3, Lemma 2.1]), while the S -segment $[-1/\tau_0, \tau_0]$ approaches the deformed-Mellin contour γ_ε along $\text{Re}(\tau) = \varepsilon$ from 0 to $i\infty$. Hence

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \lim_{\Lambda \rightarrow \infty} L^*(f, s; \gamma_{\varepsilon+i\Lambda}^S, \gamma_{\varepsilon+i\Lambda}^T) = L_f^*(s)_{\text{int}, \gamma_\varepsilon},$$

which is (54). $\square$

5.5 Wall-crossing of the period-polynomial cocycle

The wall-crossing residue formula (10) acts on the complex- s functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$. Specialised at integer $s = \ell \in \{2, \dots, k-2\}$ and assembled into the V_{k-2} moment expansion of §2, it gives the wall-crossing of the period-polynomial cocycle of f under contour deformation across an interior pole.

Corollary 5.6 (Period-polynomial wall-crossing residue). *Let f be a meromorphic modular form of weight k on Γ , and let $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T), (\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$ be two admissible contour pairs related by a single crossing of an interior pole $\tau_p \in \mathbb{H}$ of f . Denote by $C_S^f(\gamma_{\tau_0}^S), C_T^f(\gamma_{\tau_0}^T) \in V_{k-2} \otimes \mathbb{C}$ the corresponding V_{k-2} -valued cocycle values from (12). For a simple pole $\text{Res}(f, \tau_p)$,*

$$C_S^f(\gamma_{\tau_0}^S) - C_S^f(\gamma_{\tau'_0}^S) = 2\pi i (2\pi i)^{k-2} \sigma_{\tau_p}^S \text{Res}(f, \tau_p) (X - \tau_p Y)^{k-2}, \quad (55)$$

and analogously for C_T^f with $\sigma_{\tau_p}^S$ replaced by $\sigma_{\tau_p}^T$. The integer- $s = \ell$ specialisation of (10) is the coefficient of $\binom{k-2}{\ell} (-1)^\ell X^{k-2-\ell} Y^\ell$ in (55) (combined for S and T). For $\dim S_k = 1$ and $f \in F_k$ meromorphic with interior poles, the period polynomial r_f extracted via Theorem 1.1 from $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ therefore depends on the homotopy class of the contour pair, with jump (55) across each wall.

Proof. The integrand $f(\tau)(X - \tau Y)^{k-2}$ of (12) has a simple pole at $\tau = \tau_p$ with residue $\text{Res}(f, \tau_p)(X - \tau_p Y)^{k-2}$, so a contour deformation crossing τ_p with winding $\sigma_{\tau_p} \in \{-1, +1\}$ contributes $2\pi i \sigma_{\tau_p} \text{Res}(f, \tau_p)(X - \tau_p Y)^{k-2}$ to the moment integral. Multiplying by the cocycle prefactor $(2\pi i)^{k-2}$ of (12) gives (55). Expanding $(X - \tau_p Y)^{k-2}$ in the monomial basis and matching coefficients of $X^{k-2-\ell} Y^\ell$ for each $\ell \in \{0, \dots, k-2\}$ reproduces (10) at integer $s = \ell$ (with $\tau_p^{\ell-1}$ arising from $X^{k-2-\ell} Y^\ell$ at $\ell \geq 1$ from the S -segment contribution, and analogously for the T -segment via $\tilde{k}_T(\tau_p, \ell)$ at integer ℓ). $\square$

5.6 Remark on the polylog-projection machinery

The polylog projection $\hat{P}_{\tau_p} : f \mapsto \sum_{m>0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e(\tau - \tau_p))$ (see [3, §2.2]) subtracts the leading polar tails of f at each interior pole τ_p , ensuring that the Λ -truncated Mellin integral against f has a finite $\Lambda \rightarrow \infty$ limit (necessary in the deformed-Mellin construction because of the Hagedorn-rate Fourier coefficients $|c_f(n)| \sim n^{p-1} e^{2\pi n \text{Im}(\tau_p)}$ generated by interior poles [3, Prop. 1.1, 2.7]). In the finite-contour formulation (2) the $\Lambda \rightarrow \infty$ limit is not taken, so $\hat{P}_{\tau_p}$ is not required for the existence or local-constancy properties of L^* . The polylog-projection machinery does remain relevant, however, for the *closed-form per-mode evaluation* of L^* in incomplete gammas parallel to (7), where the Hagedorn-rate Fourier modes must be split off and dealt with separately before each mode contributes a clean incomplete-gamma term. For forms regular at the cusp this per-mode evaluation is available in closed form, as follows.

Proposition 5.7 (Per-mode closed form for interior-pole forms regular at the cusp). *Let $f \in F_k$ be regular at the cusp, with interior poles at the points τ_p of a fundamental domain and principal-part data $r_{f,\tau_p}^*(m)$. Then, for every admissible contour pair in the homotopy class of Corollary 1.4, $L^*(f, s)$ has the per-mode closed form*

$$\begin{aligned} L^*(f, s) = & -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right] \\ & + \sum_{\tau_p} \sum_{m > 0} r_{f,\tau_p}^*(m) [I_{m-1}(s, \tau_p) + i^k I_{m-1}(k-s, \tau_p)], \end{aligned} \quad (56)$$

where $c_{Rf}(n)$ are the Fourier coefficients of the polylog-regularised part $(1-\hat{P})f$ (with $\hat{P} = \sum_{\tau_p} \hat{P}_{\tau_p}$), and $I_N(s, \tau_p)$ are the polylog integrals of [3, §2].

We state Proposition 5.7 without reproducing its derivation: it is the $B = 1$ specialisation of the L -function of [3, §2] for $L_f^*(s)_{\text{int}}$, obtained there by the polylog projection $\hat{P}_{\tau_p}$, and is inherited here through Corollary 1.4. The genuinely new regime — f with poles *both* at interior points and at the cusp, where the $\hat{P}_{\tau_p}$ projection of [3] must be combined with the finite-contour cusp treatment of §4 — is left to the sequel.

6 Outlook

Two questions are left open. First, the geometric content of the bare Mellin kernel $f(\tau) \tau^{s-1} d\tau$: its integer- s specialisation is the polynomial V_{k-2} -kernel $(X - \tau Y)^{k-2}$, but at general s we conjecture that $\tau^{s-1} d\tau$ is a section of an infinite-dimensional principal-series local system on $Y(1)$, in the one-parameter family of the Lewis–Zagier period functions for Maass forms [13]; identifying it, and recasting the Hurwitz inversion of δ_T as a coboundary statement there, is open. Second, a systematic per-mode evaluation of L^* on interior-pole forms parallel to (7), adapting the polylog projection $\hat{P}_{\tau_p}$ of [3] to the finite-contour setting and matching the wall-crossing residues $\mathcal{R}_{\tau_p}(f, s)$ to the iterated-Eichler-integral framework of [1]; the CM cases $\tau_p \in \{i, \rho\}$, where the Laurent data is arithmetic, are a clean target. Summed over a Γ -orbit, the residues (10) assemble into a weight- k Poincaré-style sum in the pole location, taken up separately.

A Algebraic derivation of the Hurwitz-zeta kernel

The body of §4.2 derives $\tilde{k}_T(\tau, s)$ by differentiating the two-segment functional in τ_0 and demanding the result vanish. Here we record the parallel derivation that proceeds *from* the classical Mellin transform of a holomorphic cusp form, splitting it at the contour endpoints $\{-1/\tau_0, \tau_0\}$ and demanding the resulting tail be representable as a T -segment integral. Both derivations produce the same shift-difference equation and the same kernel; this appendix makes the algebraic provenance of the Hurwitz-zeta combination manifest.

Step 1: Mellin split at the contour endpoints. Let $f \in S_k$ with k even. The classical completed L -function $L^*(f, s) = (2\pi)^{-s} \Gamma(s) L(f, s)$ has integral representation

$$L^*(f, s) = -i \int_0^{i\infty} f(\tau) (\tau/i)^{s-1} d\tau = i^{-s} \int_0^{i\infty} f(\tau) \tau^{s-1} d\tau,$$

the second equality using $(\tau/i)^{s-1} = i^{1-s}\tau^{s-1}$ on the imaginary axis. Pick $\tau_0 \in \mathbb{H}$ and split the imaginary-axis contour at the two endpoints $\{-1/\tau_0, \tau_0\}$ in (1). Reorganising the three resulting pieces using S -modularity $f(-1/\tau) = \tau^k f(\tau)$ on the lower segment $[0, -1/\tau_0]$ (via the substitution $\tau \mapsto -1/\tau$) and computing the q -series-against- τ^{s-1} integral on $[\tau_0, i\infty]$ directly yields, for every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,

$$L^*(f, s) = i^{-s} \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^{s-1} d\tau + \sum_{n \geq 1} c_f(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right]. \quad (57)$$

The left side is τ_0 -independent (it is the classical L -value); both terms on the right side individually depend on τ_0 , but their sum does not. At $\tau_0 = i$ the argument $-2\pi i n \tau_0 = 2\pi n$ is real and (57) reduces to the standard BFK incomplete-gamma sum of [2].

The right side of (57) contains the S -segment integral $i^{-s} \int_{-1/\tau_0}^{\tau_0}$ explicitly; the remaining incomplete-gamma sum plays the role that we want to be a T -segment integral.

Step 2: ansatz for the T -segment kernel. Postulate that there exists a kernel $\tilde{k}_T(\tau, s)$, analytic in $\tau \in \mathbb{H}$ and meromorphic in $s \in \mathbb{C}$, such that

$$i^{-s} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = \sum_{n \geq 1} c_f(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right] \quad (58)$$

for every $f \in S_k$ and every $\tau_0 \in \mathbb{H}$. If such a kernel exists, substituting (58) into (57) gives the two-segment functional $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = i^{-s} [\int_{\gamma_{\tau_0}^S} + \int_{\gamma_{\tau_0}^T}]$ as a τ_0 -independent expression equal to $L^*(f, s)$.

Step 3: τ_0 -derivative gives a shift-difference equation. Differentiate both sides of (58) in τ_0 . On the left, the only τ_0 -dependence is in the integration limits, so $d/d\tau_0$ acts as

$$\frac{d}{d\tau_0} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = f(\tau_0) [\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s)],$$

using $f(\tau_0 - 1) = f(\tau_0)$ for the lower limit. On the right of (58), differentiating each incomplete gamma in its second argument (the explicit τ_0 -dependence) and assembling the Fourier series yields, after canceling a common factor of $f(\tau_0)$,

$$\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) = -\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s} \quad \text{for all } \tau_0 \in \mathbb{H}. \quad (59)$$

The first term on the right is the contribution of the τ^{s-1} -Mellin endpoint at τ_0 ; the second term is the S -modularity image of the same endpoint at $-1/\tau_0$, carrying the branch factor $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{-(s-1)}$ together with the modularity Jacobian τ_0^k . Equation (59) is a shift-difference equation of order one in τ , with a sum of two power-of- τ terms on the right side.

Step 4: Hurwitz raising solves the shift-difference equation. The Hurwitz zeta function $\zeta(\sigma, a)$ satisfies the raising identity

$$\zeta(\sigma, a+1) - \zeta(\sigma, a) = -a^{-\sigma} \quad (\sigma \in \mathbb{C}, \operatorname{Re}(a) > 0) \quad (60)$$

(see [6, §25.11.3]). Applied at $(\sigma, a) = (1-s, \tau)$ and $(\sigma, a) = (s-(k-1), \tau)$ respectively, this gives

$$\zeta(1-s, \tau+1) - \zeta(1-s, \tau) = -\tau^{s-1}, \quad \zeta(s-(k-1), \tau+1) - \zeta(s-(k-1), \tau) = -\tau^{k-1-s}.$$

Combining with the relative phase $e^{i\pi(s-1)}$ from the second term of (59) gives the kernel

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1), \quad (61)$$

which by construction satisfies (59) and hence (58). The kernel is analytic in τ on $\mathbb{H}$ (both Hurwitz zetas have second argument $\tau+1$ with $\text{Re}(\tau+1) > 0$) and meromorphic in s with simple poles only at $s=0$ and $s=k$, where the per-mode integrals collect removable cancellations (see Proposition 4.4). Equation (61) is the kernel (3) of the main text.

Where DR2022 stops. The kernel (61) has two Hurwitz-zeta terms. The construction of [8, Thm. 1.2] gives the single-term

$$\tilde{k}_T^{\text{DR}}(\tau, m) = \zeta^*(1-m, \tau+1)$$

on the segment $[i, i+1]$, at integer m , which is the first term of (61). The structural origin of this gap is the absence of an S -segment in their construction: without the S -modularity-transformed contribution at the $-1/\tau_0$ endpoint in Step 3, the shift-difference equation (59) has only the single right-hand-side term $-\tau_0^{s-1}$, and its solution is the single-Hurwitz kernel. The second $e^{i\pi(s-1)} \tau_0^{k-1-s}$ term appears precisely because the two-segment functional includes the S -segment and its endpoint at $-1/\tau_0$ contributes under τ_0 -differentiation via S -modularity.

B Eichler–Shimura for physicists

This appendix is intended as an exposition aid for readers approaching the subject from theoretical physics. It is removable without affecting the logical content of the paper and is excluded from the journal/arXiv version. Primary references: Brown [1] §2–3 (clean modern exposition), Kohnen–Zagier 1984 [12] (classical), and Lewis–Zagier [13] (accessible motivation).

B.1 The polynomial coefficient system V_n and the slash action

Fix k even, set $n := k-2$, and let V_n be the $\mathbb{Q}$ -vector space of polynomials in two formal variables X, Y , homogeneous of total degree n . $\Gamma = \text{SL}_2(\mathbb{Z})$ acts on V_n on the right by the *slash action*

$$(P|_\gamma)(X, Y) := P(aX + bY, cX + dY), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \quad (62)$$

The two generators of Γ act as $P|_T(X, Y) = P(X+Y, Y)$ and $P|_S(X, Y) = P(-Y, X)$ on the basis monomials $X^{n-\ell}Y^\ell$.

B.2 Group cohomology $H^1(\Gamma; V_n)$

A 1-cocycle of Γ with coefficients in V_n is a map $C : \Gamma \rightarrow V_n$ satisfying

$$C_{gh} = C_g|_h + C_h, \quad g, h \in \Gamma. \quad (63)$$

This is exactly the property derived in Lemma 2.1. A 1-coboundary is one of the form $C_\gamma = P|_\gamma - P$ for some fixed $P \in V_n$. Coboundaries are automatically 1-cocycles. The *first group cohomology* is

$$H^1(\Gamma; V_n) := \{1\text{-cocycles}\} / \{1\text{-coboundaries}\}. \quad (64)$$

Because $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ is generated by S and T subject to the relations $S^2 = -I$, $(ST)^3 = -I$, a 1-cocycle is determined by the pair $(C_S, C_T) \in V_n \oplus V_n$ subject to the algebraic constraints

$$C_S|_S + C_S = 0, \quad C_T + C_{ST}|_{ST} + C_{ST}|_S = 0, \quad (65)$$

the latter coming from $(ST)^3 = -I$ together with $C_{(ST)^3} = C_{ST}|_{(ST)^2} + C_{ST}|_{ST} + C_{ST}$. Coboundaries $P|_\gamma - P$ contribute the pair $(P|_S - P, P|_T - P)$.

B.3 The Eichler integral cocycle

For $f \in S_k^!$ and basepoint $\tau_0 \in \mathbb{H}$, set

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^n d\tau \in V_n \otimes \mathbb{C}. \quad (66)$$

This is the *Eichler integral cocycle* of f ; it is a 1-cocycle by Lemma 2.1. Its cohomology class $[C^f] \in H^1(\Gamma; V_n \otimes \mathbb{C})$ depends on f but not on τ_0 (for f without interior poles), because changing τ_0 adds a coboundary by (16).

B.4 The Eichler–Shimura theorem

The cohomology decomposes canonically as

$$H^1(\Gamma; V_n \otimes \mathbb{C}) \cong H_{\mathrm{cusp}}^1(\Gamma; V_n \otimes \mathbb{C}) \oplus H_{\mathrm{Eis}}^1(\Gamma; V_n \otimes \mathbb{C}), \quad (67)$$

with the cuspidal summand isomorphic to $S_k \oplus \overline{S_k}$ via the map $f \mapsto [C^f]_{\mathrm{cusp}}$. In the dim $S_k = 1$ substantive family $k \in \{12, 16, 18, 20, 22, 26\}$, the cuspidal summand is 2-dimensional, spanned by the two *cuspidal basis polynomials* $p_f^+, p_f^- \in V_n \otimes \mathbb{C}$ [1]. These are characterised by their behaviour under the S -action: p_f^+ satisfies $p_f^+|_S = p_f^+$ (even parity) and p_f^- satisfies $p_f^-|_S = -p_f^-$ (odd parity). Concretely, S acts on V_n by $(X, Y) \mapsto (-Y, X)$, so for $P = \sum_\ell p_\ell X^{n-\ell} Y^\ell$,

$$P|_S = \sum_\ell p_\ell (-Y)^{n-\ell} X^\ell = \sum_\ell (-1)^{n-\ell} p_\ell X^\ell Y^{n-\ell}; \quad (68)$$

that is, S interchanges $X \leftrightarrow Y$ and signs flip according to the monomial. The $\pm$ parity is then the action of this involution on $V_n \otimes \mathbb{C}$.

The Eisenstein summand H_{Eis}^1 is also non-trivial; for $\dim S_k = 1$, it is spanned by the Eisenstein coboundary class represented (up to scale) by the coboundary $\delta Y^n(\gamma) = Y^n|_\gamma - Y^n$. At $\gamma = S$, this gives $(-X)^n - Y^n = X^n - Y^n$ (using n even), the polynomial that contributes only to the boundary monomials X^n, Y^n of any representative cocycle.

B.5 Periods $\omega^\pm(f)$ and quasi-periods $\eta^\pm(f)$

For $f \in S_k$ (genuine holomorphic cusp form), the cuspidal projection of $[C^f]$ in the (p_f^+, p_f^-) basis defines two complex scalars,

$$[C^f]_{\mathrm{cusp}} = \omega^+(f) p_f^+ + \omega^-(f) p_f^-, \quad (69)$$

called the *periods* of f . By the classical Manin/Eichler/Shimura identity, these encode the critical L -values $L(f, \ell) = \int_0^\infty t^{\ell-1} f(it) dt$ for integer ℓ in the critical strip $\{1, \dots, k-1\}$.

For $f \in S_k^!$ with a cusp pole (e.g., $\hat{\Delta} = q^{-1} + O(q^2) \in S_{12}^!$), the Eichler integral C^f is still a well-defined cocycle on a finite contour, and the cuspidal projection $[C^f]_{\mathrm{cusp}} = \omega^+(f) p_f^+ + \omega^-(f) p_f^-$ defines a pair of scalars by the same formula. In this weakly holomorphic case they are denoted $\eta^\pm(f)$ and called *quasi-periods*; their relation to L -values requires the finite-contour regularisation studied in the main note (Theorem 1.1).

B.6 The Eisenstein-direction term in r_f versus the period polynomial of an Eisenstein series

A subtlety worth stating directly, in response to a common confusion. Two different objects are sometimes both called “period polynomials of Eisenstein origin”, and they have very different properties.

Object A: The Eisenstein direction in H^1 . In the decomposition (29) of r_f into cuspidal-basis polynomials $p_f^\pm$ plus an “Eisenstein-direction” term, the latter is by construction a multiple of the coboundary representative $Y^n|_S - Y^n = X^n - Y^n$ (or, equivalently, the polynomial $X^n - Y^n$ itself, which spans the Eisenstein direction in H^1). This contribution touches *only* the boundary monomials X^n and Y^n of r_f , leaving all interior monomials $X^{n-(\ell-1)}Y^{\ell-1}$, $\ell \in \{2, \dots, k-2\}$, free of Eisenstein contamination. This is what makes the interior-monomial extraction of $\omega^\pm(f)$ via (4) clean.

Object B: The period polynomial of an actual Eisenstein series. The period polynomial $r_{E_k}(X, Y)$ of the weight- k Eisenstein series $E_k(\tau)$ is a different object: it is the formal S -cocycle value of C^{E_k} modulo coboundaries, computed for $f = E_k$ itself. By Manin/Zagier (see [12] §2), r_{E_k} is supported on *all* monomials $X^\ell Y^{n-\ell}$ for $\ell \in \{0, 1, \dots, n\}$, with coefficients given by products of critical zeta values:

$$r_{E_k}(X, Y) \sim \sum_{\ell=0}^n \zeta^*(\ell+1) \zeta^*(k-1-\ell) \binom{n}{\ell} X^\ell Y^{n-\ell}, \quad (70)$$

where $\zeta^*(s) := \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is the completed Riemann zeta. The factorisation $L(E_k, s) = \zeta(s) \zeta(s-k+1)$ sends *some* interior coefficients to zero — precisely those at ℓ with $\zeta(\ell-k+2) = 0$, i.e., $\ell-k+2 \in \{-2, -4, -6, \dots\}$ (trivial zeros of ζ) — but *not* all interior coefficients. For $k=12$, the vanishing happens at $\ell \in \{3, 5, 7, 9\}$ (and the parity-paired $\ell \in \{8, 6, 4, 2\}$ remain nonzero, modulo sign-pairing details). So $r_{E_{12}}$ is *not* supported only at the boundary monomials $\ell = 0, n$.

The distinction. Object A is a statement about *the structure of H^1* : the Eisenstein direction projects onto a 1-dimensional subspace whose representative coboundary touches only boundary monomials. Object B is a statement about *a specific cocycle representative* (the period polynomial of E_k), which generically has interior support. The two are not in conflict because what we actually use in the proof of Theorem 1.1 is Object A: the Eisenstein *direction’s* contribution to r_f at interior monomials vanishes, regardless of what the period polynomial of an actual Eisenstein series looks like.

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- [4] Companion `mpmath` verification scripts (50-digit precision). `verify_periods.py` checks Theorem 1.1 at $k = 12$ for $f \in \{\Delta, \hat{\Delta}\}$: it reproduces Brown’s $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ [1], confirms basepoint independence and the closed form for δ_T^{-1} (Lemma 3.2), and verifies the kernel identity of Lemma 3.5 symbolically. `dr_b_Lfunction_complex_s.py` checks Theorem 1.2 at non-integer s (half-integer, off-axis, and outside-strip points), reproducing $L^*(\Delta, s)$ to $\sim 10^{-53}$ and $L^*(\hat{\Delta}, s)$ to $\sim 10^{-47}$, with τ_0 -independence. `verify_s6.py` and `verify_multi_s.py` check the per-mode $\tau_0 = i$ form of Theorem 1.2 at $s \in \{1, 3, 6, 9, 11\}$, including the boundaries $s = 1, 11$ and the polar mode $n = -1$.
- [5] F. Brown, personal communication to the author, 28 June 2019. This correspondence describes the two-segment cocycle procedure (integrate along contours from $\gamma^{-1}\tau_0$ to τ_0 for the two generators $\gamma \in \{S, T\}$ of Γ , modify by a coboundary $P|(1 - T)$ to kill the T -component, then read $\omega^\pm$ and $\eta^\pm$ off the S -component), and is the structural starting point of the construction in the present note. An accompanying unpublished note on quasi-periods of cusp forms of levels 6 and 8 was attached to the correspondence.
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